

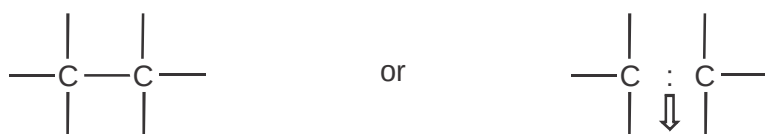
UNIT-7

Basic Concepts in Organic Chemistry and Stereochemistry

1. ELECTRONIC EFFECTS

1. Inductive Effects : Permanent displacement of sigma electrons along the carbon chain when an atom or group of atoms having different electronegativity is attached with this chain is called inductive effect.

When a covalent bond is present between two atoms which have same electronegativity, the shared electron pair lie exactly between the two atoms.



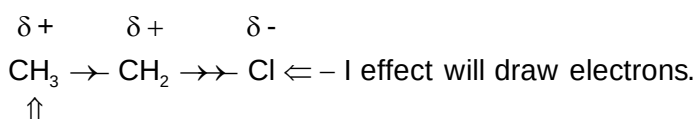
- Covalent bond
- Electronegativity of both atom same.
- Equal distributions of electrons
- Electron pair occupied central position
- Non polar covalent bond

In a covalent bond between unlike atoms, the electron pair forming the sigma bond is never shared equally. It tends to be attached a little more towards the more electronegative atom. As a result of this polarity develops in the molecule.



The characteristics of the inductive effect are as follows :

- (1) It is a permanent effect.
- (2) Inductive effect is distance dependent electronic factor. The effect weakens steadily with increasing distance from the substituent.
- (3) The inductive effect depends on the electronegativity of the substituent.
- (4) The inductive effect is transmitted through sigma bond.



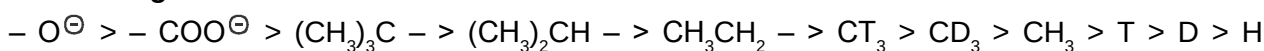
A slightly positive charge on the carbon atom

Types of Inductive Effect

(A) + IE (+I effect)

Groups or atoms which are less electronegative than hydrogen atom are said to have +IE

Decreasing Order of +IE :



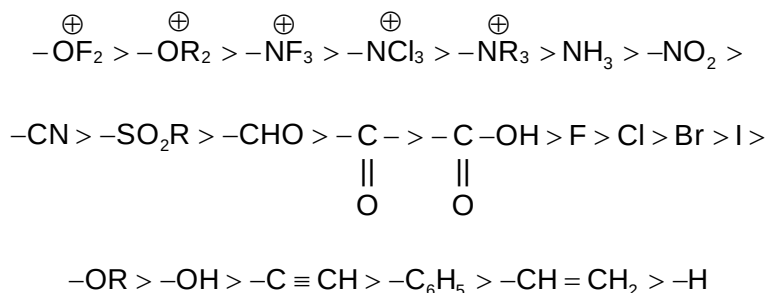
All alkyl groups have +IE

Havier the isotope, more will be the +IE

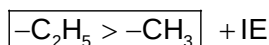
(B) -IE (-I effect)

Groups or atoms which are more electronegative than hydrogen atom are said to have -IE

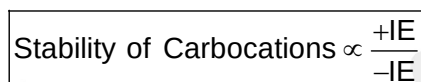
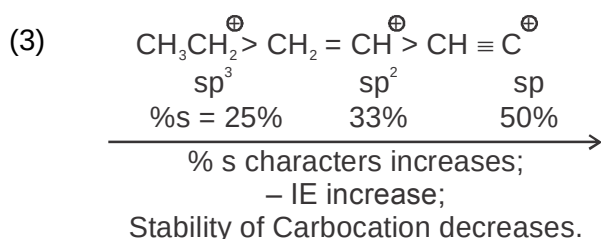
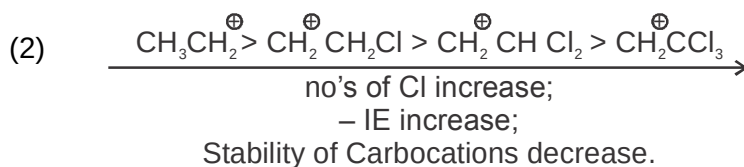
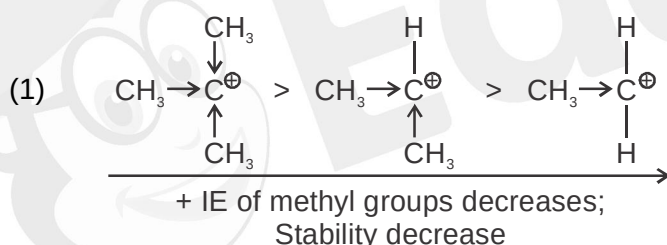
Decreasing Order of -IE :



Greater the number of carbon atoms of the alkyl group greater is the +IE

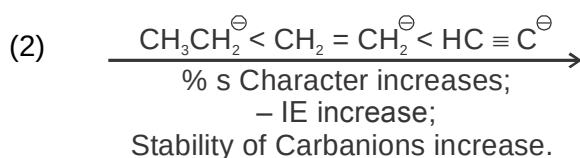
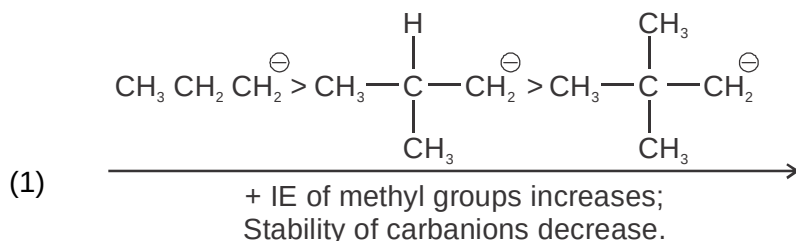

Applications of Inductive Effect

- (1) Stability of Intermediates
- (2) Acidic nature of compounds
- (3) Basic nature of Amines

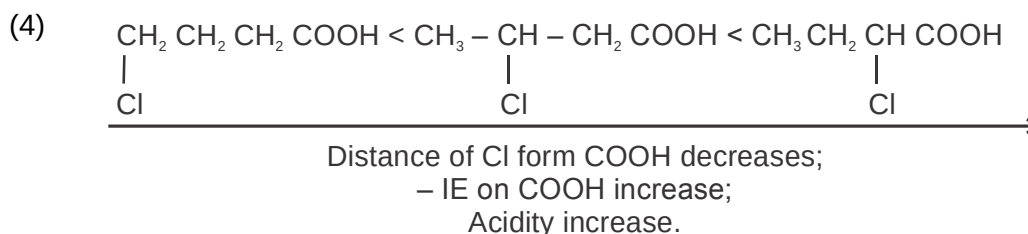
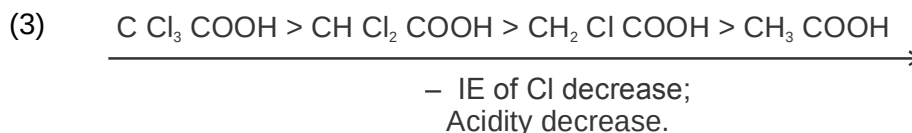
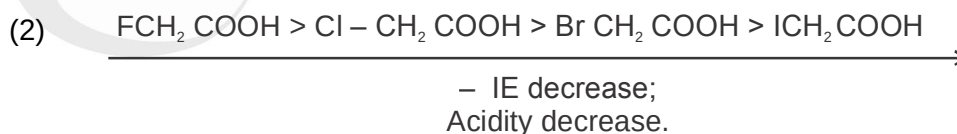
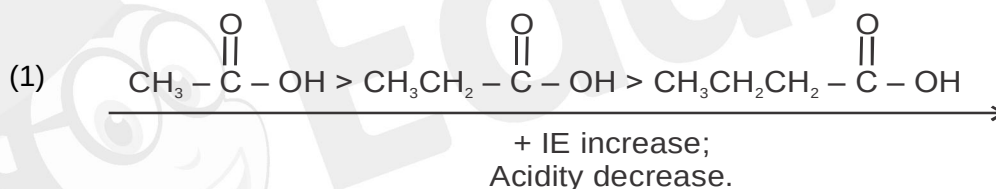
(1) Stability of Intermediates
(A) Stability of Carbocations :

Examples :


(B) Stability of Carbanions :

$$\text{Stability of Carbanions} \propto \frac{-\text{IE}}{+\text{IE}}$$

Examples :

(2) Acidic nature of Compounds :

$$\text{Acidity} \propto \frac{-\text{IE}}{+\text{IE}}$$

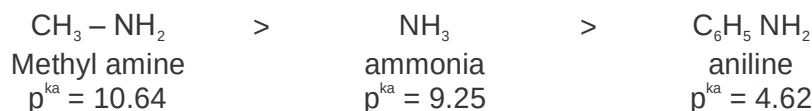
Examples :


(3) Basic Nature of Amines

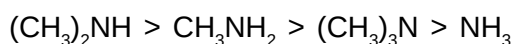
Basic nature of amines depends upon lone pair availability on N atom which it can donate.

When one or more hydrogen atom(s) of ammonia are replaced by the alkyl groups, then the electron pair availability on N atom increases due to + IE of alkyl group.

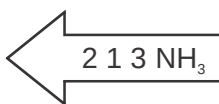
But in case of aromatic amines due to - I effect of phenyl group, basicity of aromatic amines increase in comparison to ammonia.



In aqueous solution; the basicity of amines are



Therefore, the order of basicity can be easily remembered by the formula

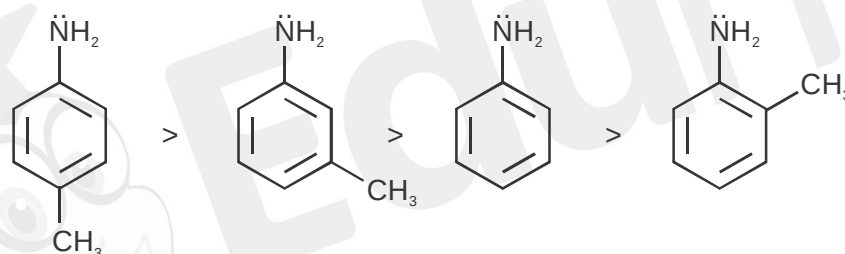


but in the gas phase, the basicity of amines are

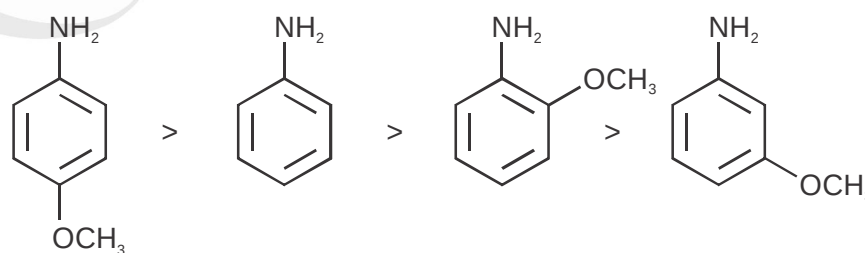


Basicity of Aromatic Amines

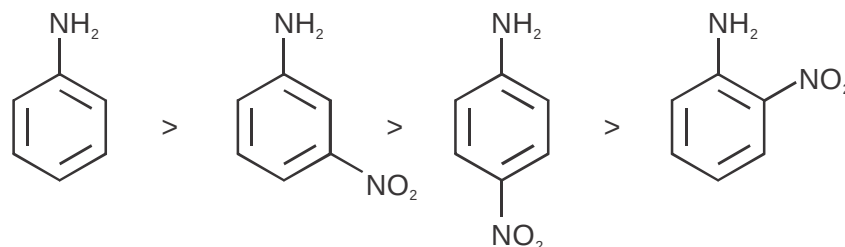
Case-I : When alkyl group attached to benzene ring, the order of basicity is as follows :



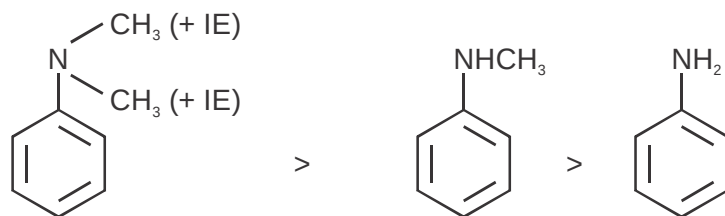
Case-II : When methoxy group is attached to the benzene ring, the order of basicity is :



Case-III : When a nitro group is attached to the benzene ring, the order of basicity is :



Case-IV : In the case of N-alkyl substituted aryl amines, the order of basicity is :



Resonance or Mesomeric Effect

The resonance effect (Mesomeric effect) involves delocalisation of electrons through resonance via the π -system.

Resonance : Two or more structures of the same molecule each differing in the distribution of electrons.

The properties of these structures will not be those to be expected of any of the resonating structures, but they will be those to be expected of a combination called a resonance hybrid of them. The phenomenon is called resonance and represented by the symbol $\leftrightarrow$

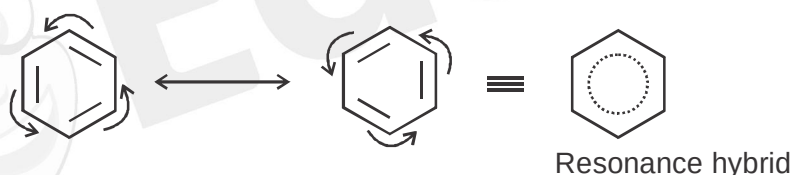
The resonance hybrid is the real structure of the chemical species.

For example,

We can not explain all the properties of Benzene if we consider a single structure of benzene i.e

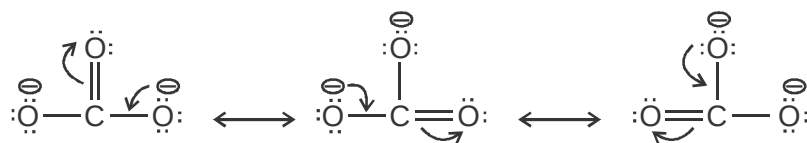


However, if we consider following two kekule structures for benzene we can explain all the properties of benzene.

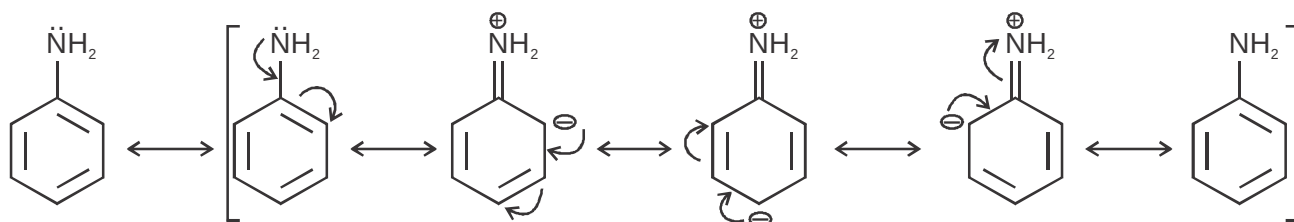


Some Examples of Resonance are -

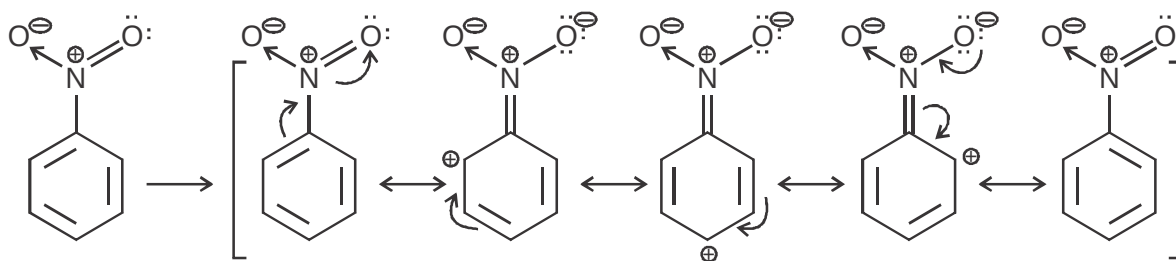
(i) Carbonate Ion



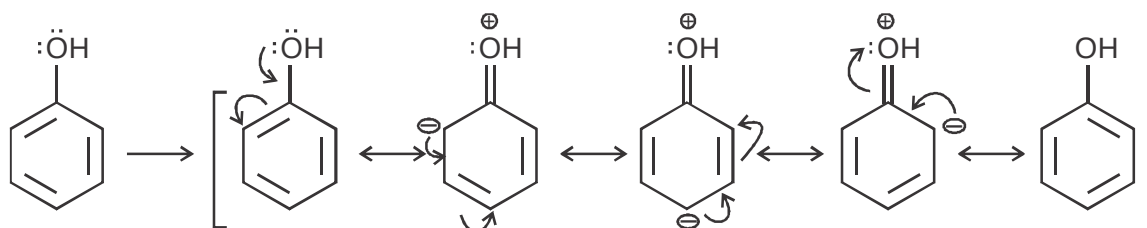
(ii) Aniline



(iii) Nitrobenzene

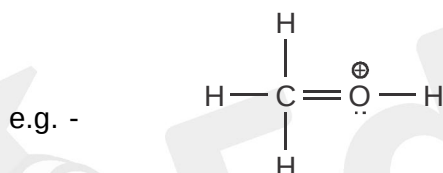


(iv) Phenol



Rules for Resonance Contribution of a Resonating Structures

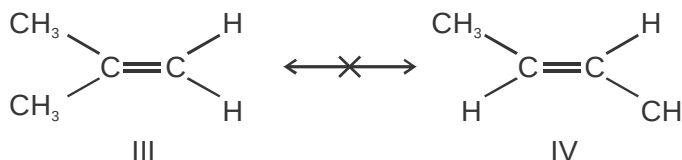
- (1) All the canonical structures (Resonating structures) must be written according to the lewis method showing bonds, lone pair of electrons, and formal charge.



- (2) The position of the nuclei in each of the canonical structures must be in the same relative position. Only electrons movement is allowed.

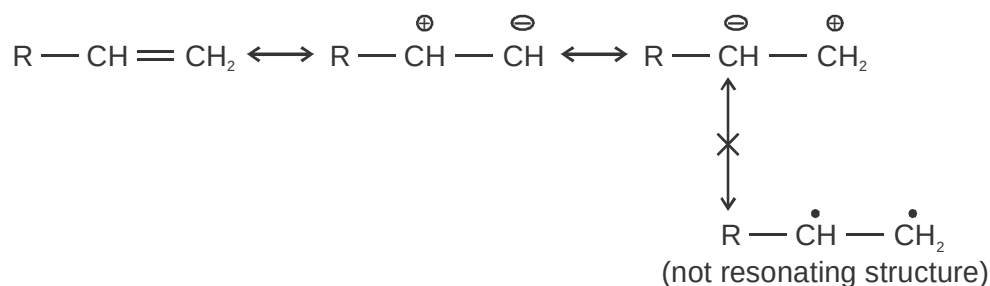


Thus I and II are canonical structures



III and IV are not resonating structures because H atom has been moved.

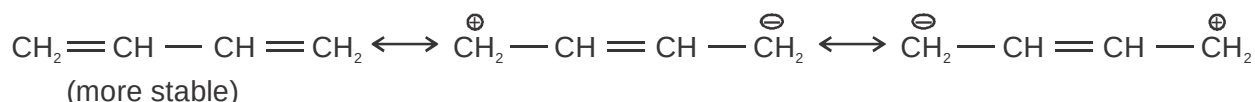
- (3) All the resonating structures must have the same number of unpaired electrons.



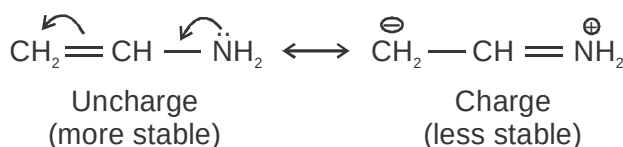
- (4) All the atoms taking part in resonance must lie in same plane or nearly in the same plane so that maximum overlapping of p-orbital can be possible.
- (5) The energy of actual molecule is lower than the energy of any contributing structure.
- (6) All the resonating structures do not contribute equally to the resonance hybrid. The more stable structure is the great contribution.

The stabilities of resonating structures :

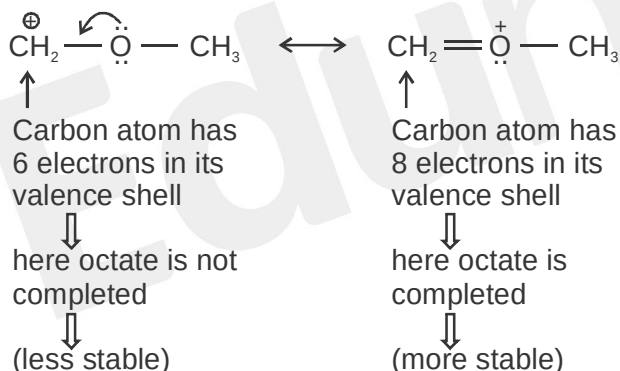
- (1) Resonating structures with maximum numbers of covalent bonds are normally more stable e.g. :



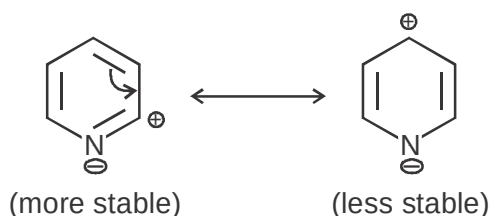
- (2) If different resonating structures have the same number of covalent bond, then uncharged structures are more stable. e.g. :



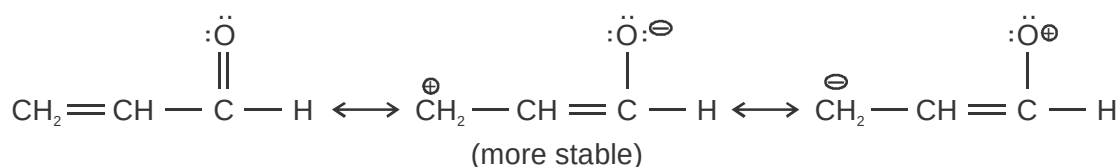
- (3) Structures in which all of the atoms have a complete octate are preferred. e.g. :



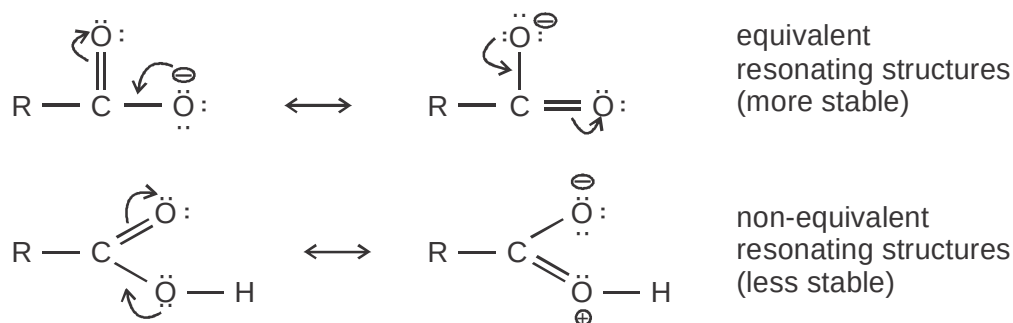
- (4) Structures in which there is maximum distance between like charges or minimum distance between unlike charges, are more stable. e.g. :



- (5) Structures with negative formal charge is present on a more electronegative atom and positive charge on the least electronegative atom is more stable. e.g. :

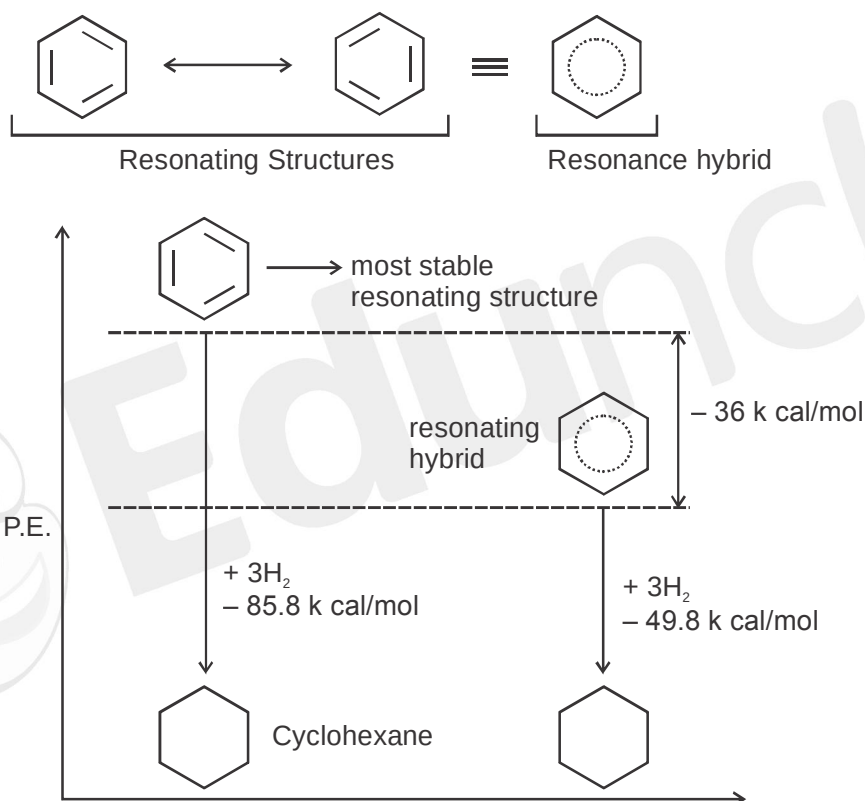


- (6) Equivalent resonating structures are more stable than non equivalent resonating structures.
e.g. :



Resonance Energy

The resonance energy is a measured of the extra stability of the resonance hybrid. The difference in energy between the actual structure (resonance hybrid) and the most stable resonating structure is called resonance energy. e.g. :



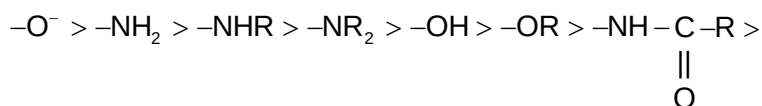
Thus benzene has 36 k cal/mol less energy than that of the most stable resonating structure expected for a typical compound with three double bonds. Hence resonance energy of benzene is 36 k cal/mol.

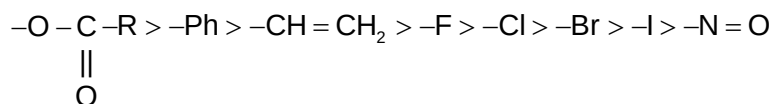
Types of Mesomeric Effect :

There are two types

(i) + M Effect : The capacity of a group to donate electrons into a molecular π -system is described as a +M effect.

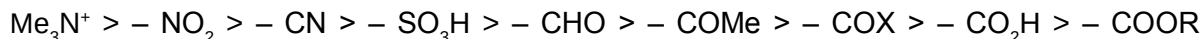
Groups which shows '+M' effect are :





(ii) – **M Effect** : The capacity of a group to withdraws electrons from a molecular π -system is described as a –M effect.

Groups which shows '–M' effect' are :



Applications of Resonance

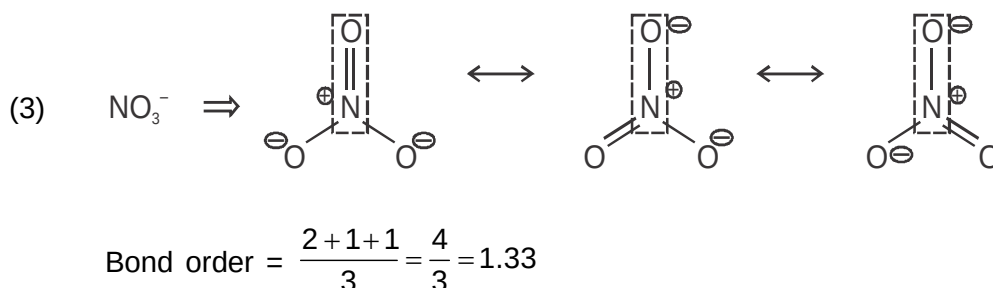
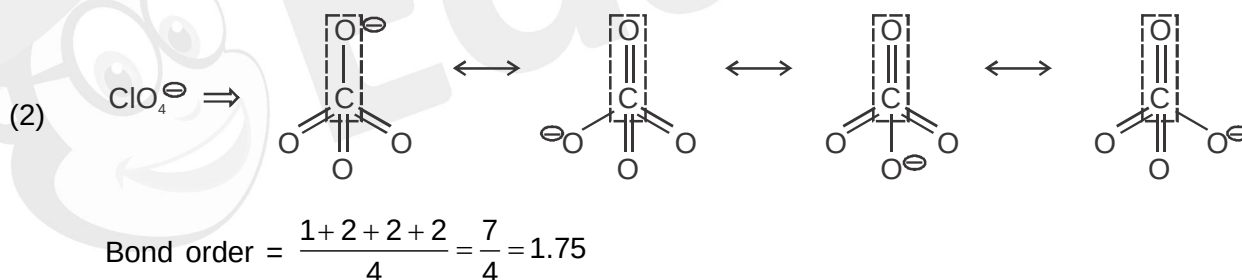
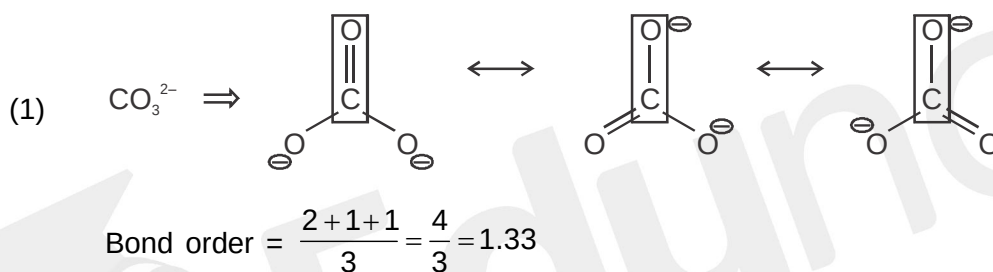
(1) Stability of Intermediate

Resonance has been used to explain stability of the reaction intermediate like carbocations, carbanions, carbene free radicals etc.

(2) Bond Order

$\text{Bond order} = \frac{\text{Total no's of covalent bond between two bonded atom is all resonating structures}}{\text{Total resonating structures}}$
--

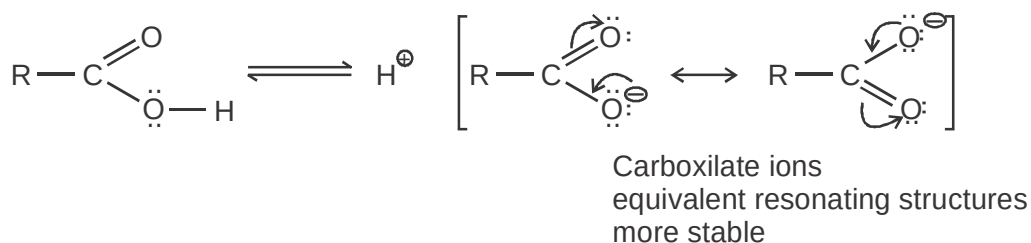
For example,



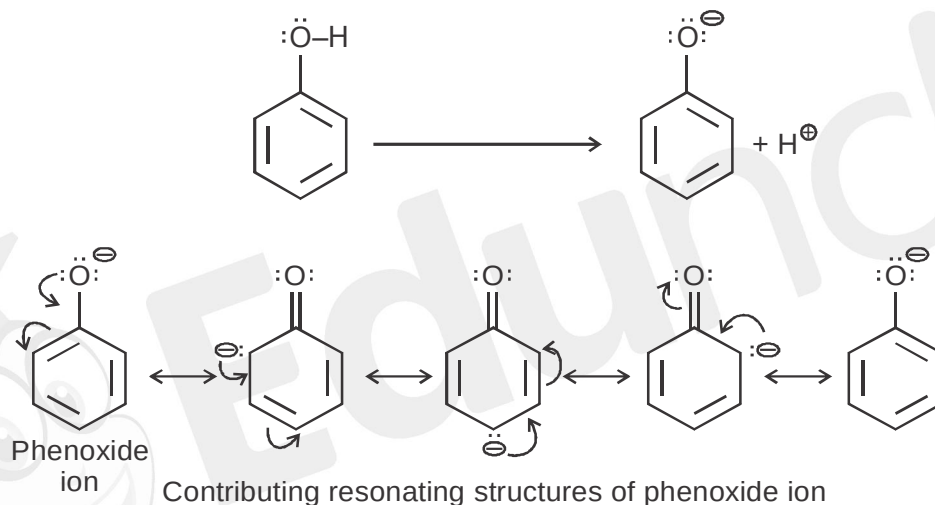
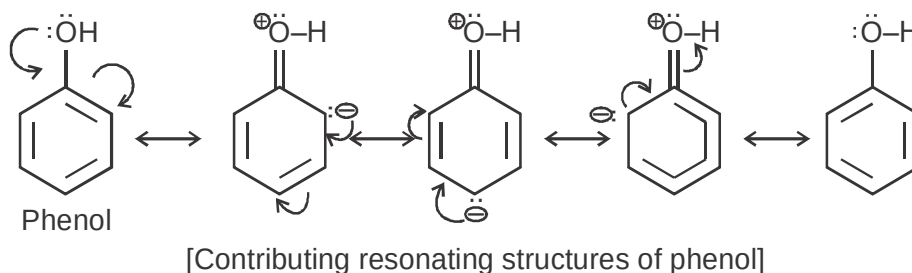
(3) Acidity and Basicity

(1) **Acidity of Carboxylic Acids** : Carboxylic acid is acidic due to the resonance stabilization of the carboxylate ion.

Carboxylate ions exist as two equivalent resonating structures and moreover the negative charge is shared.



(2) Acidity of Phenols : Phenol itself is resonance stabilized and formed resonance stabilized anion namely phenoxide anion.

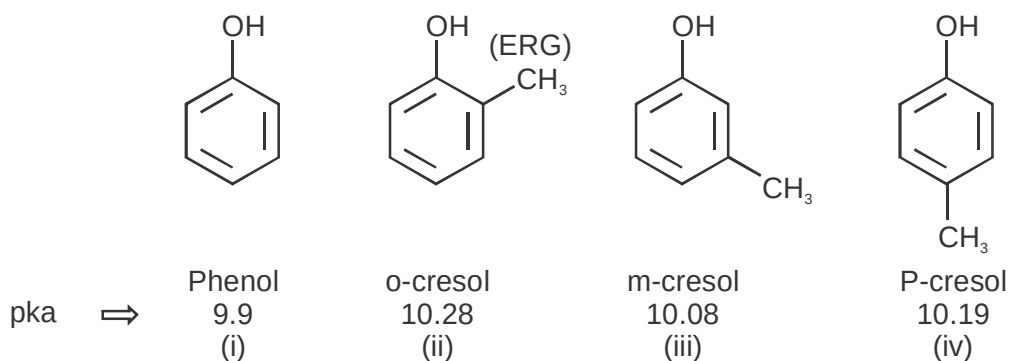


If electron withdrawing group (EWG) is attached to the benzene ring, it increase the acidic nature of phenol.

If electron releasing group (ERG) is attached to the benzene ring, it decrease the acidic nature of phenol.

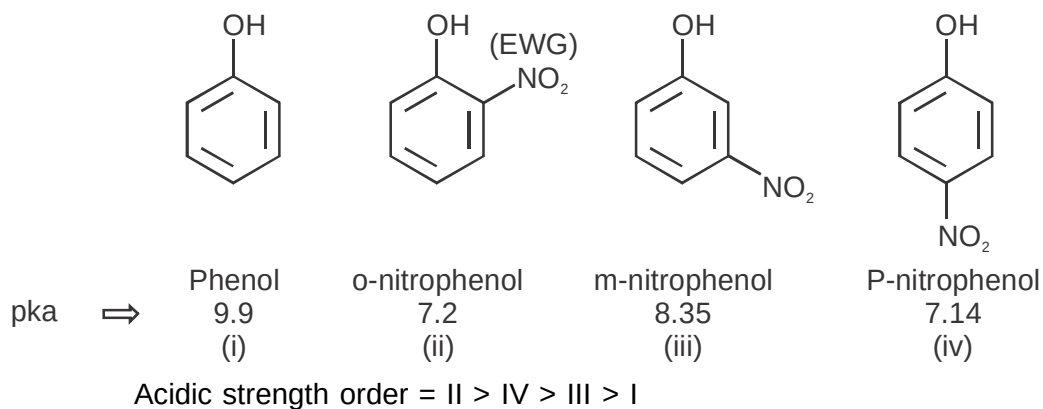
Examples :

(1)

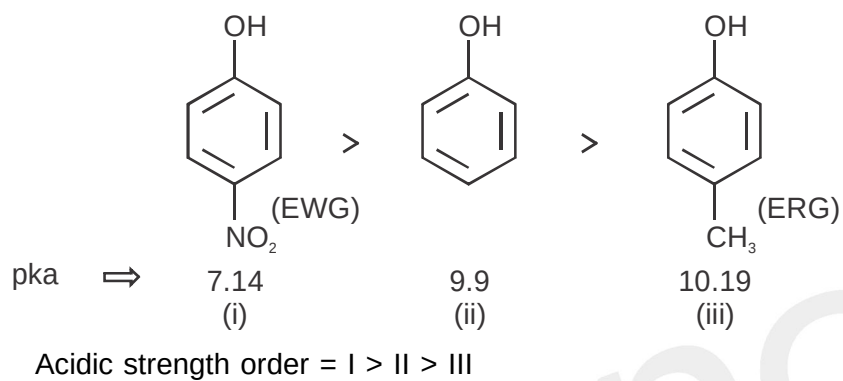


Acidic strength order = I > III > IV > II

(2)

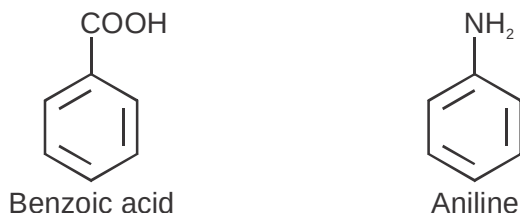


(3)

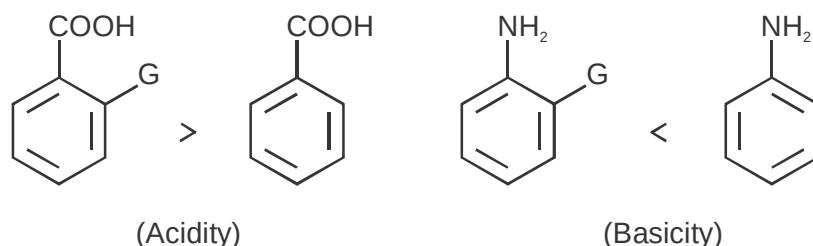


2. STERIC EFFECTS

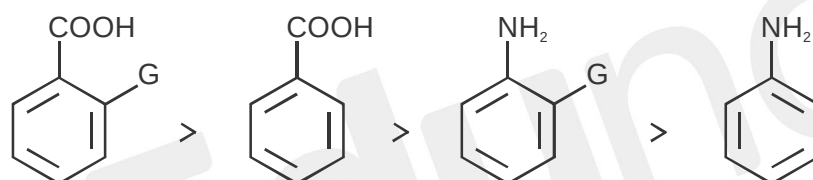
(1) **Ortho Effect** : Ortho effect is observe in benzoic acids and Anilines



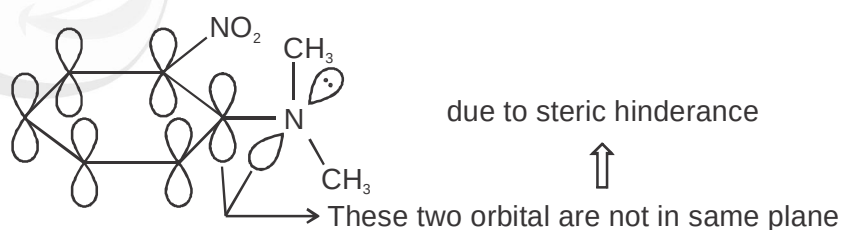
Ortho substituted benzoic acids are highly acidic than benzoic acid.
While orthosubstituted Anilines are weaker base than anilines.



Acidity Order :



(2) **Steric inhibition of Resonance (SIR Effect)** : The most important condition for resonance to occurs is that the resonating structures must be coplanar. There are many examples in which resonance is inhibited because orbitals are sterically forced out of planarity. This phenomenon is called steric inhibition of resonance.



Aromaticity

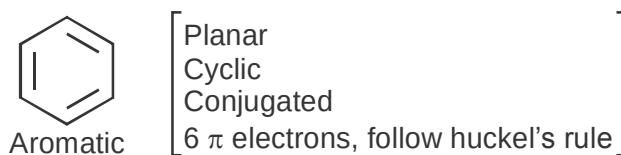
The term 'Aromaticity' may be defined as the ability of an organic compound like benzene to undergo substitution a rather than addition reaction.

A Compound is said to be aromatic if it fulfils the following conditions -

- (1) The structure must be cyclic having conjugated π -bonds.
- (2) The ring must be planar.
- (3) It must obey huckel's $(4n + 2)$ π electrons rule. It states, "if in a conjugated planar cyclic polyene, the number of delocalized π electrons is $(4n + 2)$ where n is an integer.
- (4) Each C atom of the ring must be sp^2 hybridized (or occasionally sp hybridized)
- (5) It must undergo substitution reaction.

Aromatic, Anti-aromatic and Non-Aromatic Compound

Cyclic conjugated polyenes which contain $(4n + 2)$ electrons ($n = 0, 1, 2, 3, \dots$) are aromatic
e.g. -

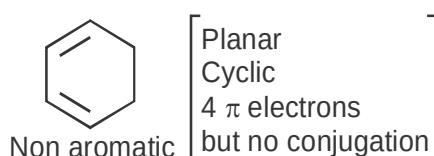


In contrast a conjugated cyclic system with 4π electrons is called antiaromatic.



A cyclic compounds in which continuous overlapping of p-orbitals disturbed cannot be aromatic or antiaromatic then it is termed as 'non aromatic'.

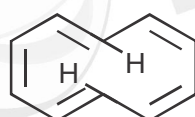
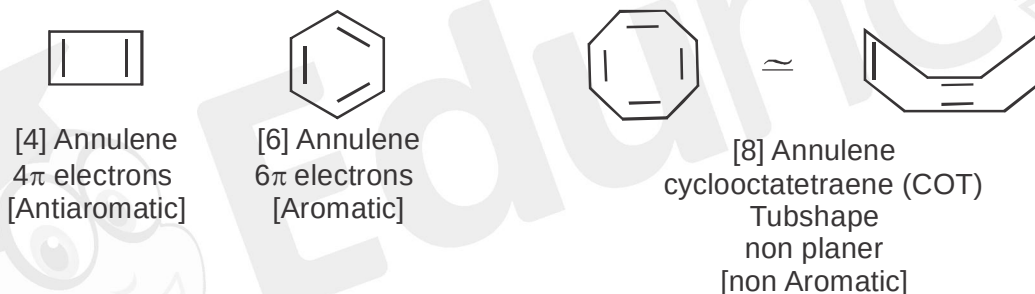
e.g. -



Examples :

(1) Annulenes : The monocyclic conjugated system of the general formula C_nH_n have been called annulenes.

The ring size of an annulene is indicated by a number in brackets.

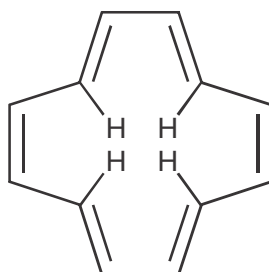


[10] Annulene
[Non Aromatic]
due to strain
Caused by the
inside hydrogens

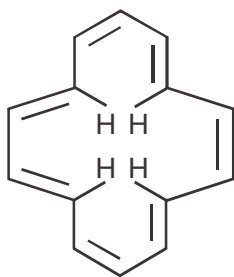


[10] Annulene
10 π electrons
Bridge cycodecapentaene
[Aromatic]

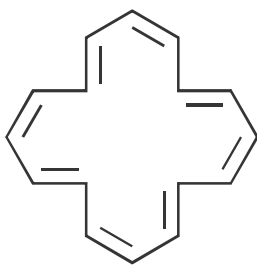
(X = CH_2 , O, NH)



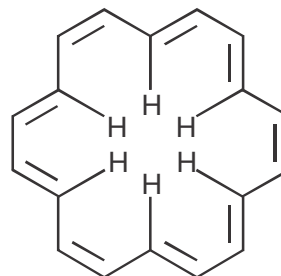
[12] Annulene
[Non Aromatic]
due to strain caused by
the inside hydrogens



[14] Annulene
14 π electrons
[Aromatic]

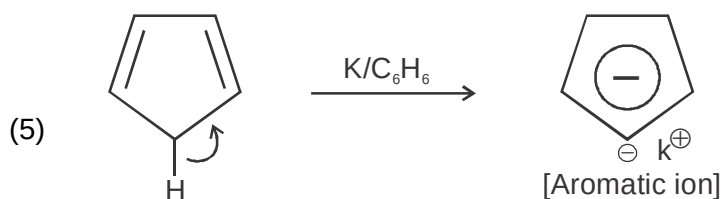
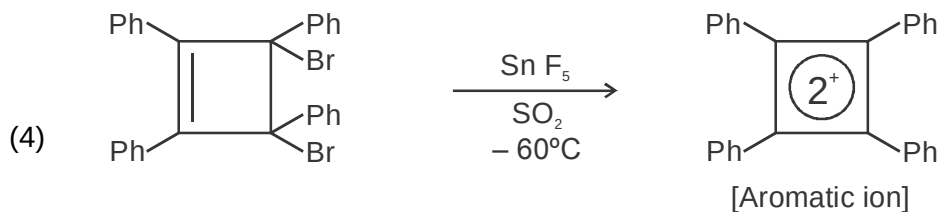
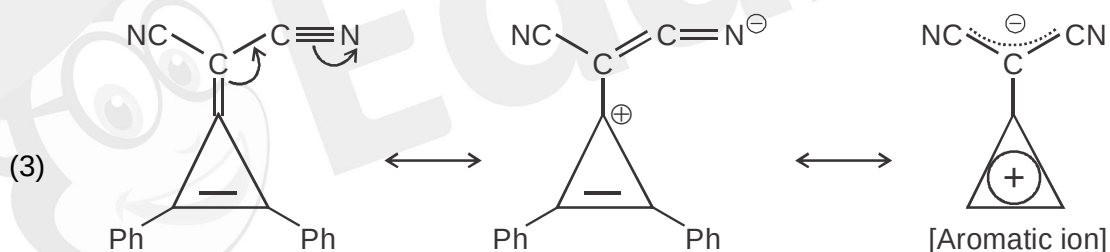
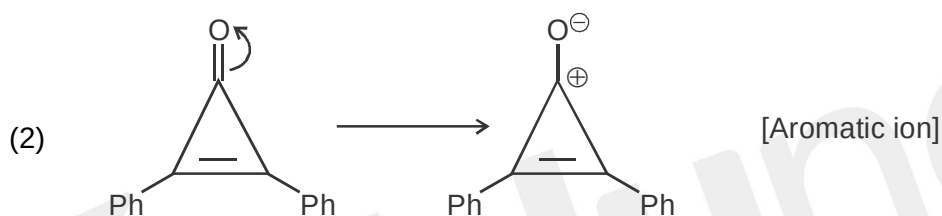
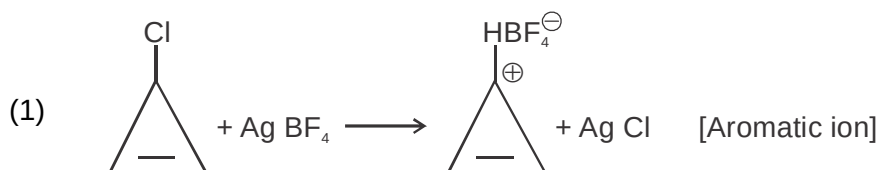


[16] Annulene
16 π electrons
non planer
[Non-Aromatic]

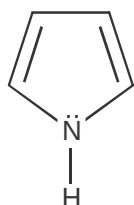


[18] Annulene
18 π electrons
[Aromatic]

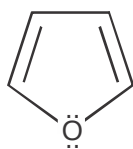
(2) Aromatic Ions :



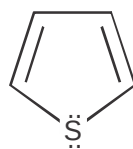
(3) Heterocyclic Compounds :



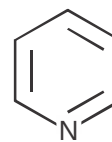
Pyrrol



Furan

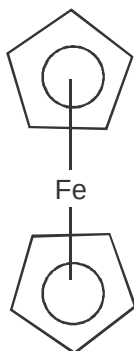
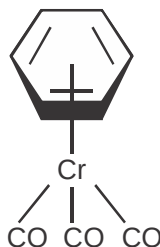


Thiophene

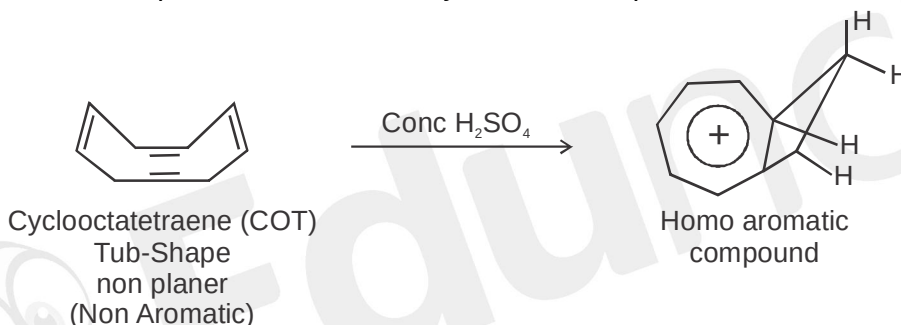


Pyridine

(4) Metallocenes :


 (Ferrocene)
Aromatic

 Benzenetricarbonylchromium
Aromatic

(5) Homoaromatic Compounds : Compounds that contain one or more sp^3 hybrid carbon are known as homoaromatic compounds and aromaticity of these compounds are known as homoaromaticity.



When COT is dissolved in concentrated sulphuric acid, a proton adds to one of the double bonds to form a homotropylium ion.

STEREOCHEMISTRY

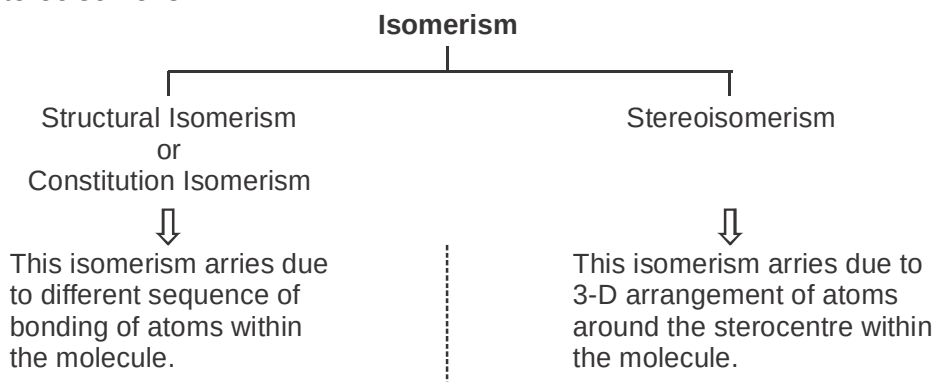
The branch of chemistry which deals with three-dimensional structures of molecules and their effect on physical and chemical properties is known as stereochemistry.

Introduction

Compounds that have the same molecular formula, same molecular weight but different physical and chemical properties are called 'Isomers' and the phenomenon is known as 'Isomerism'.

There are two types of isomerism

1. Structural isomerism
2. Stereoisomerism

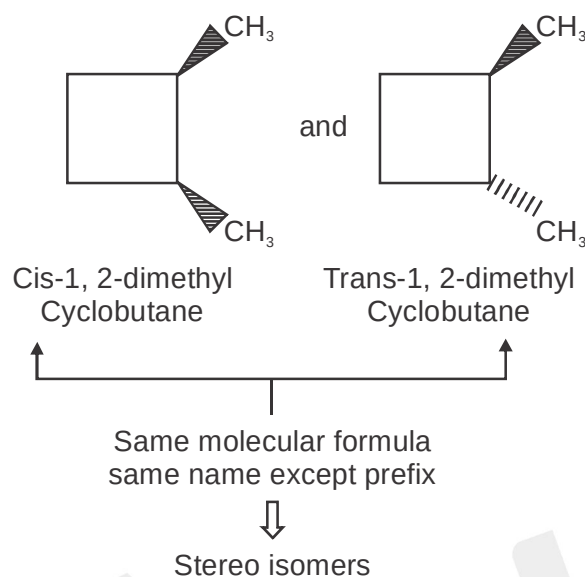


3. STEREOISOMERISM

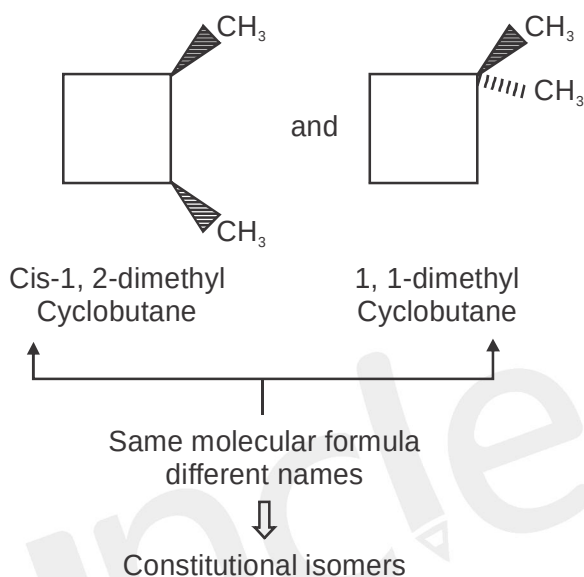
Conditions

1. Stereoisomerism differ only in the way that atoms are oriented in space.
2. Stereoisomer have identical IUPAC names (except prefix like cis and trans)
3. Stereoisomers have the same functional group.

Ex.1 C_6H_{12}



Ex.2 C_6H_{12}



It has been further classified in two categories.

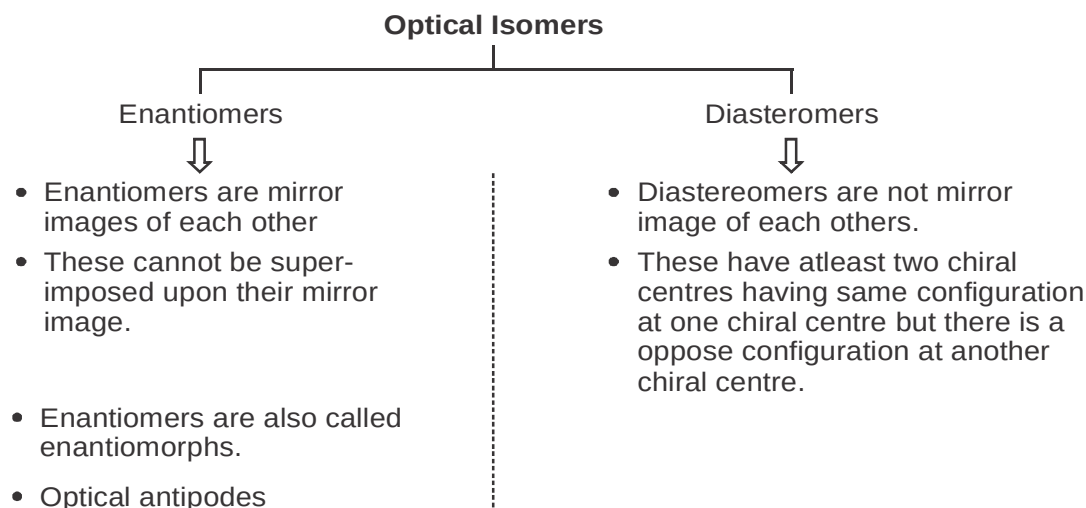
1. Geometrical isomerism
2. Optical isomerism

Optical Isomerism :

Any substances which are capable of rotating the plane of polarized light are called optical isomers and the phenomenon is called optical isomerism.

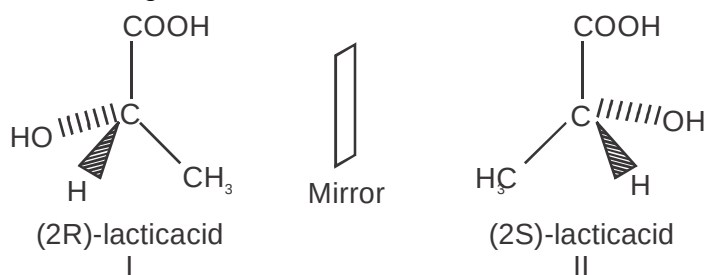
Optical isomers are two types -

1. Enantiomers
2. Diastereomers



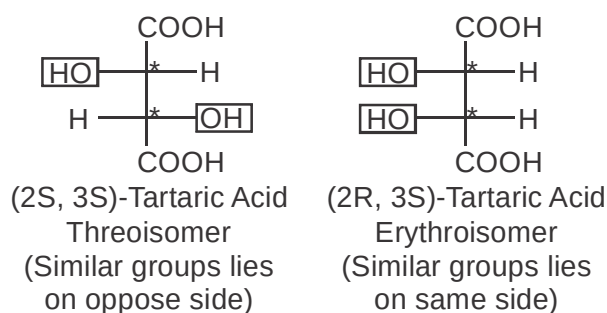
Ex.(i) (Enantiomers)

Mirror-image isomers.



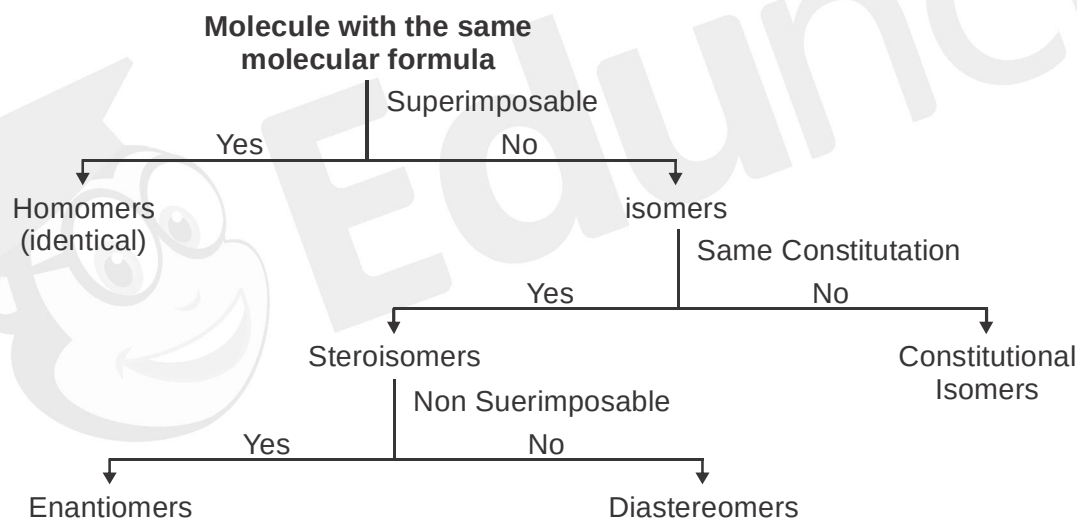
I and II are non-super imposable mirror images isomers.

Ex.(ii) (Diastereomers)



Cis-trans isomers, Anomers, Epimers are Diastereomers.

Relationship between the Molecules -



Optical Activity : The rotation of plane polarized light is known as optical activity and substances possessing this properties are called optical active. The optical activity is observed and measured by an instrument called polarimeter.

Plane Polarized Light (PPL) : In ordinary light vibration occurs in all the planes perpendicular to the direction in which light travel. When ordinary light passed through Nicol prism. Then vibration takes place in only one direction.

It is called plane polarized light (PPL)



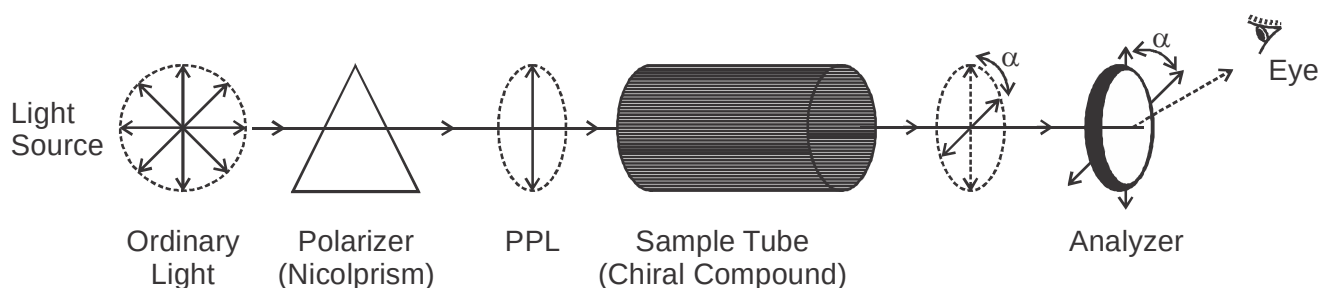


Figure : Polarimeter
Solid line : before rotation
Broken line : after rotation
 α the angle of rotation

Angle of Rotation and Specific Rotation

Optically active compounds rotate the plane of the polarized light through an angle of α .

The angle α , measured in degree ($^{\circ}$) is called the angle of rotation (α)

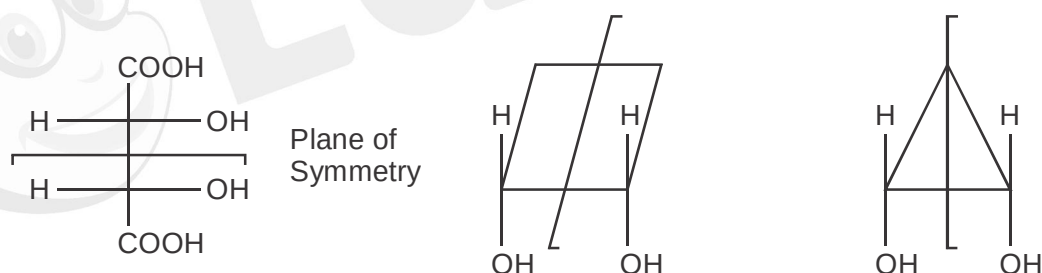
A compound that rotate the plane of polarized light is said to be optically active.

Optically active compound can exist in four forms :

- Dextrorotatory (d) or (+) form** : If the rotation of the plane is to the right (clockwise) then the substance is called dextrorotatory (Latin : Dexter = right).
- Laevorotatory (l) or (-) form** : If the rotation of the plane is to the left (Anticlockwise) then the substance is called laevorotatory. (Latin : laevus = left)
- Mesoform** : Optical isomers which have atleast two chiral centre but posses plane of symmetry is called Mesoform.

It is optically inactive due to internal compensation i.e. the rotation caused by the upper half of the molecule is nautralized by the lower half of the molecule.

Examples :



- Racemic mixture (dl) or (±)** : Equimolar mixture of Pairs of enantiomers is called racemic mixture. It is optically inactive due to external compensation i.e. the rotation caused by the d form is neutralized by the l form.

Angle of rotation depends upon

- The nature of substrate
- Concentration of the sample (c)
- Length of tube (ℓ)
- Wavelength of the light
- Temperature (t)

The common polarimeter use the sodium D-line or Mercury green line. Now new modern spectra-polarimeter use photocells because they give more accurate value.

Specific Rotation : Specific rotation is the observe angle of rotation if 1-dm (10 cm) tube is used and concentration of sample is 1g/ml.

$$[\alpha_{SP}]_D^t = \frac{\alpha}{\ell \times c}$$

$$\text{Specific rotation} = \frac{\text{observed angle of rotation } (^{\circ})}{\text{length (dm)} \times \text{g/ml}}$$

If pure sample of optically active substance is used then

$$[\alpha_{SP}]_D^t = \frac{\alpha}{\ell \times d}$$

here d represents density of a pure sample

That means specific rotation is independent on

1. Length of tube (ℓ)
2. Concentration of the sample (c)

Enantiomeric Excess : Some times we deals with a mixture that is neither optical pure enantiomer nor a racemic mixture. One enantiomer is present in excess of the other.

The enantiomeric excess (ee) also called the optical purity, tells how much more there is of one enantiomer.

$$\text{Enantiomeric excess (ee)} = \% \text{ of one enantiomer} - \% \text{ of the other enantiomer}$$

Ex.1 If a mixture contain 75% of one enantiomer and 25% of the other. Then the enantiomeric excess is ?

Sol. $ee = 75\% - 25\% = 50\%$

There is a 50% excess of one enantiomer over the racemic mixture.

Ex.2 If the enantiomeric excess is 75% then how much of each enantiomer is present?

Sol. 75% ee means that the solution contains an excess of 75% of an enantiomer and 25% of the racemic mixture of both enantiomers.

Because racemic mixture contains an equal amount of both enantiomers, it has 12.5 % of each enantiomer.

Total amount of one enantiomer = $75 + 12.5 = 87.5\%$

Total amount of other enantiomer = 12.5%

$$\begin{aligned} \% \text{ major enantiomer} &= \frac{100 + ee}{2} \\ \% \text{ minor enantiomer} &= \frac{100 - ee}{2} \end{aligned}$$

The enantiomeric excess can also be calculated as -

$$ee = \frac{[\alpha]_{\text{mixture}}}{[\alpha]_{\text{pure enantiomer}}} \times 100\%$$

$[\alpha]$ = specific rotation

$$ee = \frac{(\text{moles of major enantiomer}) - (\text{moles of minor enantiomer})}{\text{moles of both enantiomer}} \times 100$$

Conditions for Optical Activity : A compound is said to be optical active if it is non- superimposable on its mirror image due to asymmetry.

Chirality is the necessary and sufficient condition for a molecule to be optically active.

1. All Asymmetric and dissymmetric molecules are chiral and optical active.

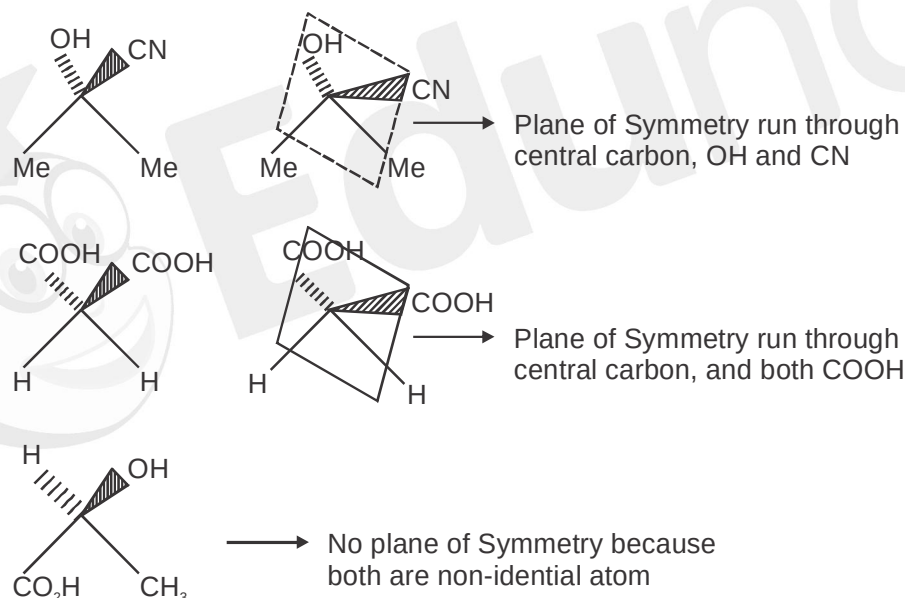
An Asymmetric molecule has no element of symmetry.

Elements of Symmetry	Asymmetric	Dissymmetric	Symmetric
1. Centre of Symmetry (i)	Absent	Absent	i-Present or σ -Present or Both Present
2. Plane of Symmetry (σ)	Absent	Absent	
3. Axis of Symmetry (C_n)	Absent	Present	
	Chiral ↓ Optically Active		Achiral ↓ Optically Inactive

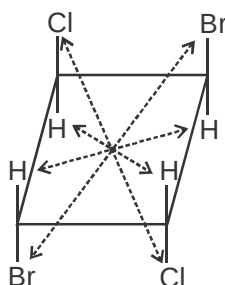
Elements of Symmetry

1. Plane of Symmetry : The plane which cuts through the middle of a molecule in such a way that one half of the molecule is a mirror image of the other half.

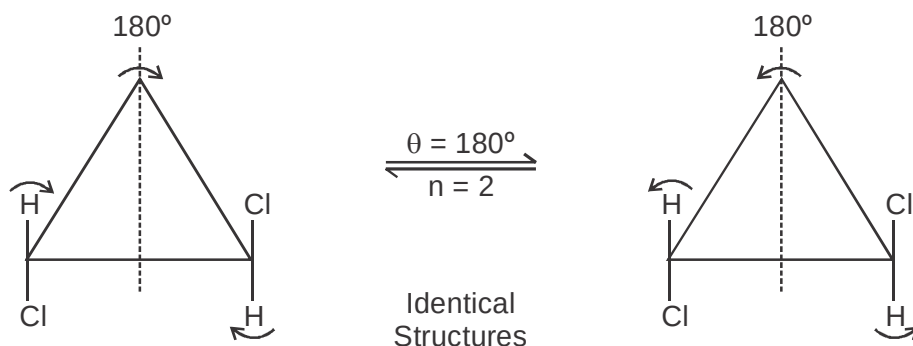
Example



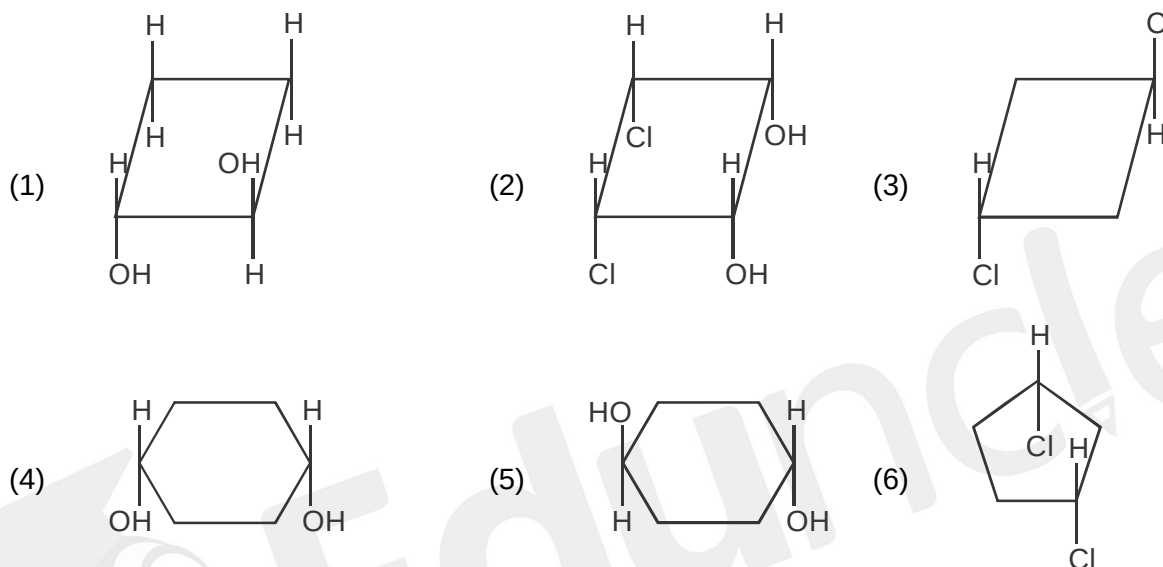
2. Centre of Symmetry(i) : A point in the molecule through if a line is drawn in one direction, then extended to equal distance in opposite direction. It meets another similar group or atom.



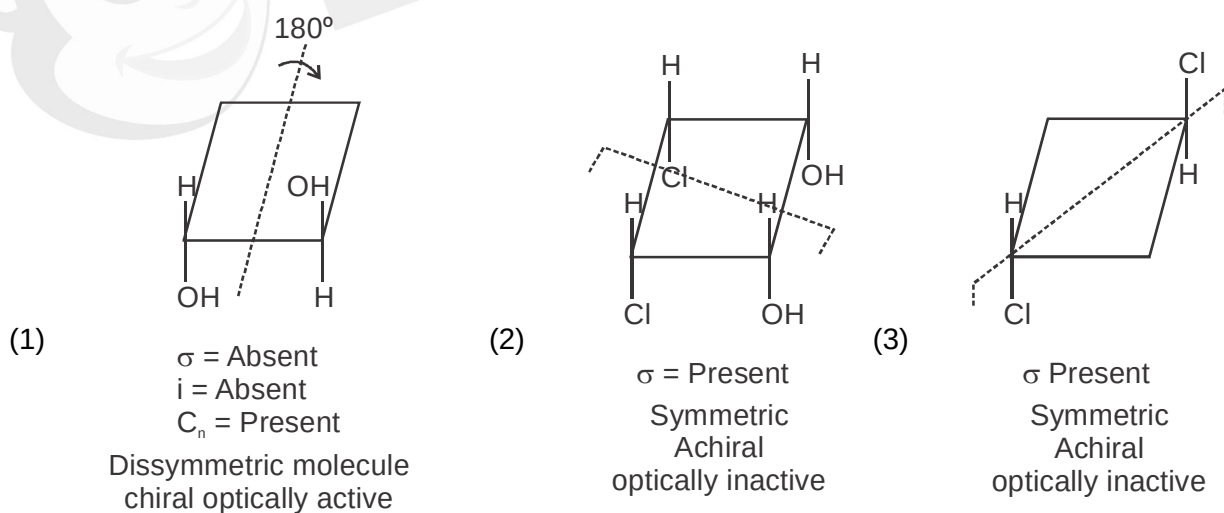
3. Axis of Symmetry (C_n) : An imaginary axis through which the object is rotated by an angle, leave the orientation of the object indistinguishable from its original orientation.

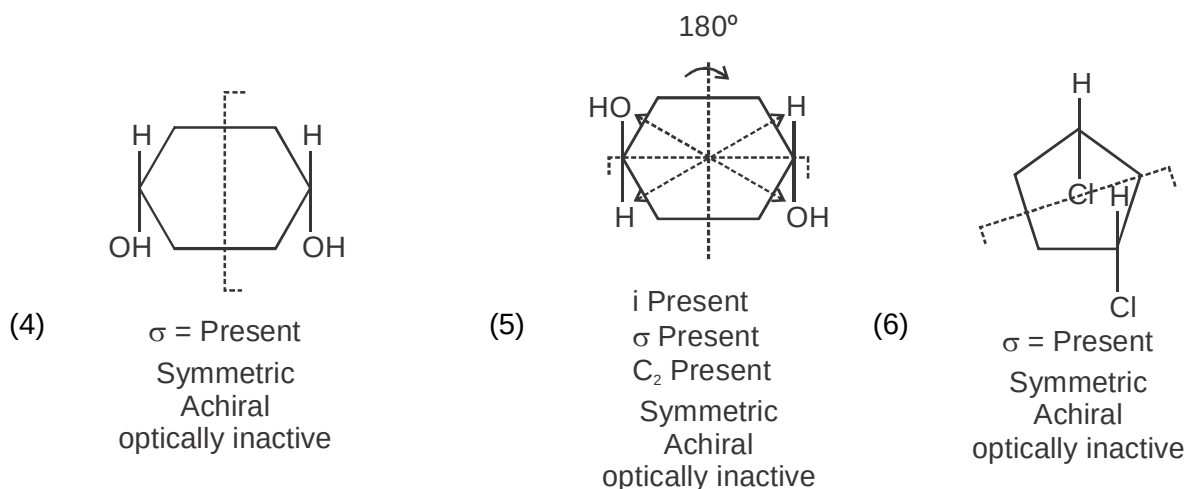


Ex. Which of the following molecules are chiral ?



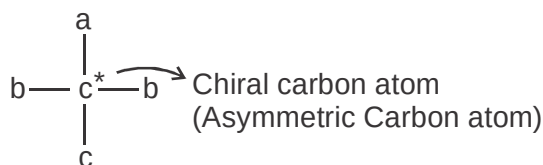
Ans. (1)
Sol.



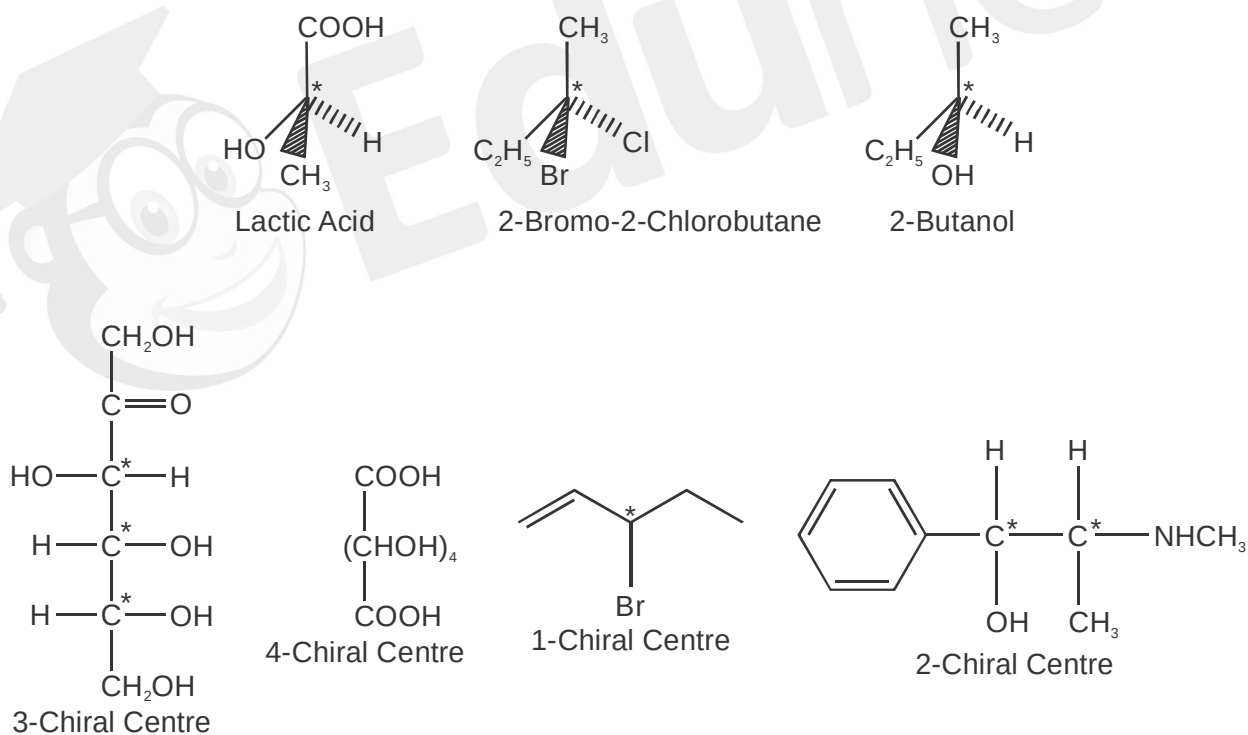


2. Compounds with Asymmetric Carbon Atom

A carbon atom is asymmetric if four different groups or atoms are attached to it



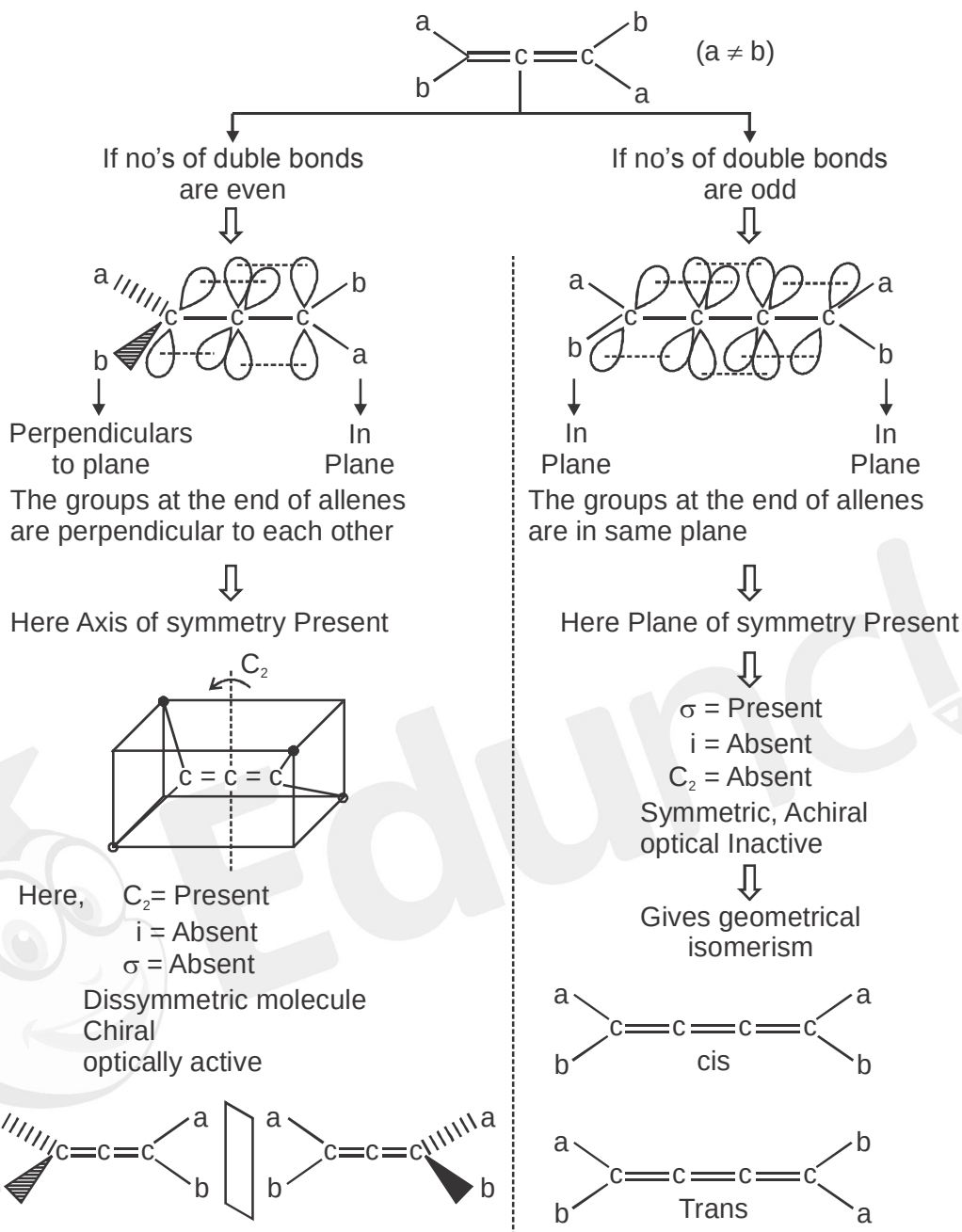
Example :



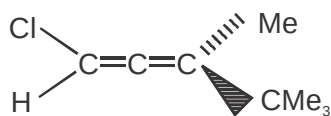
3. Optical Activity Even in The Absence of Chiral Carbon Atom

A. Optical Activity Due to a Chiral Axis :

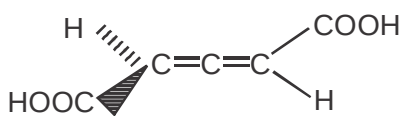
1. Allene :



Thus Allenes of type $abc = c = cab$ and $abc = c = ccd$ are chiral, when it has even no's of double bond.



1-chloro-3, 4, 4-trimethyl-
1-2-pentadiene



(R)-(-)-gutinic acid

2. Spirans :



Since the replacement of a double bond in allene by a ring does not alter the basic geometry of the molecule thus the conditions for Chirality in spirans are quite analogous to those in allenes.

here,

C_2 = Present

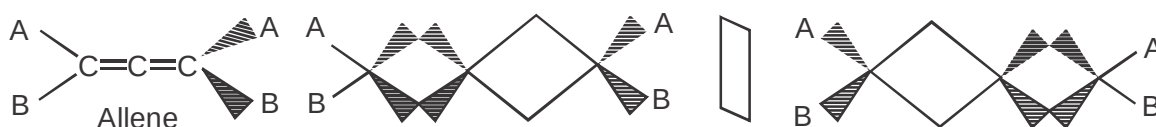
i = Absent

σ = Absent

Dissymmetric molecule

Chiral

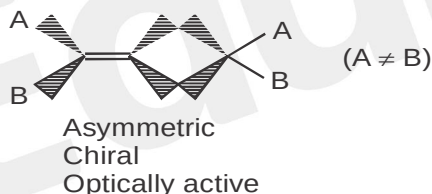
Optically active.



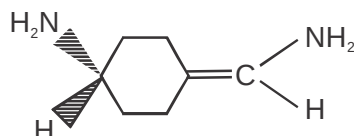
Example :



3. Hemispirans :

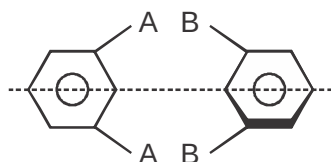


Example :

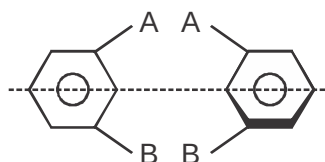


4. Biphenyls : Biphenyls show enantiomerism when the following two conditions are satisfied.

(i) Neither of the rings should possess plane of symmetry (σ)



Both rings have plane
of symmetry achiral
optically inactive

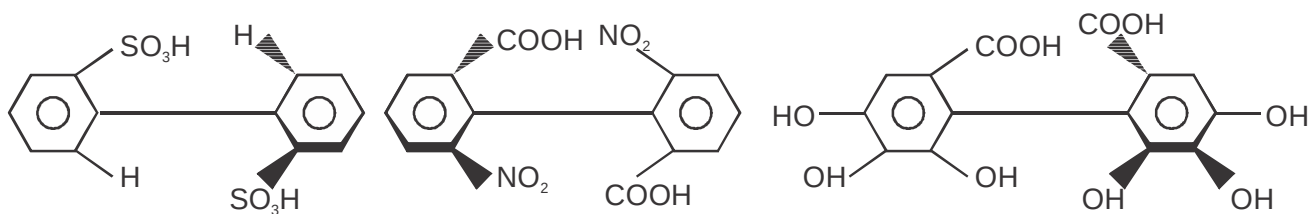


Chiral Axis
Chiral
optically active

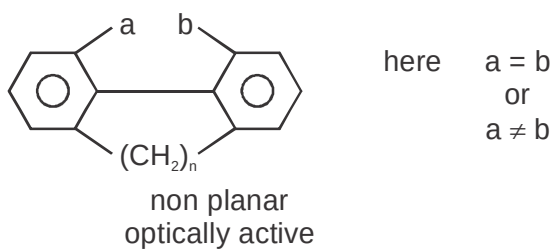
(ii) The ortho-substituents should be larger in size so that the two rings must not be coplanar, they should be orthogonal to each other

Here A and B = COOH, NO₂, I, Br etc.

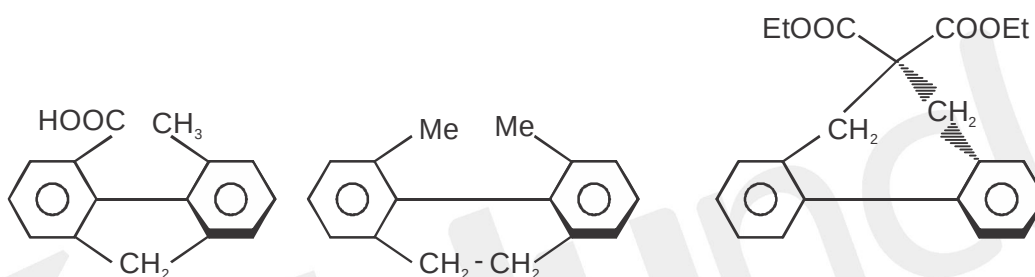
Example :



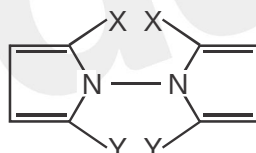
5. Bridged Biphenyls :



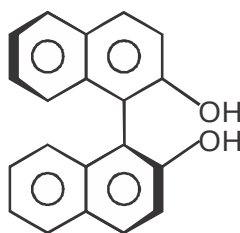
Example :



6. Bipyrrroles :

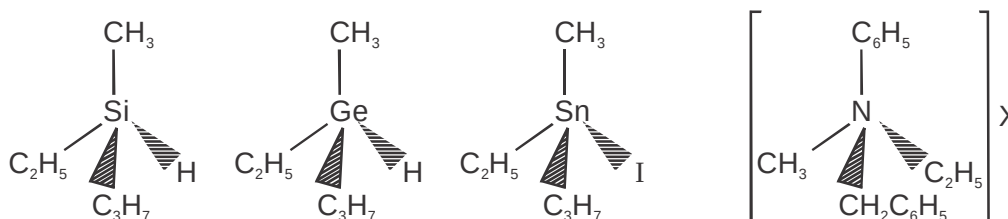


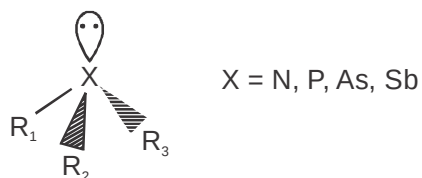
7. Binaphthols :



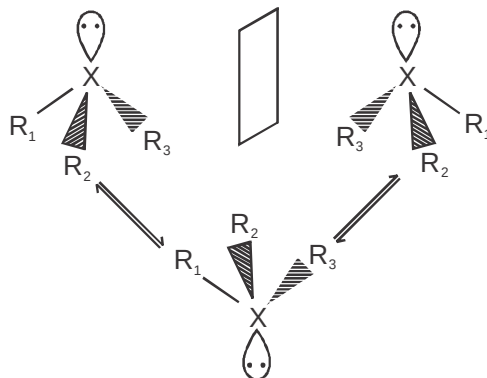
8. Compounds having chiral centres other than carbon.

Example :





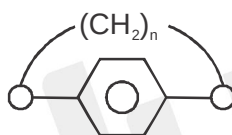
Here pyramidal inversion takes place at room temperature.



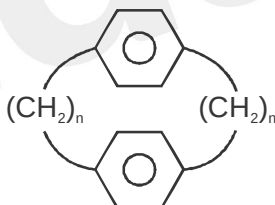
Due to pyramidal inversion both forms are interconvertible therefore they always exist as racemic mixture it is chiral but optically inactive.

B. Optical Activity Due to Chiral Plane

1. Ansa compounds :



2. Para cyclophanes :

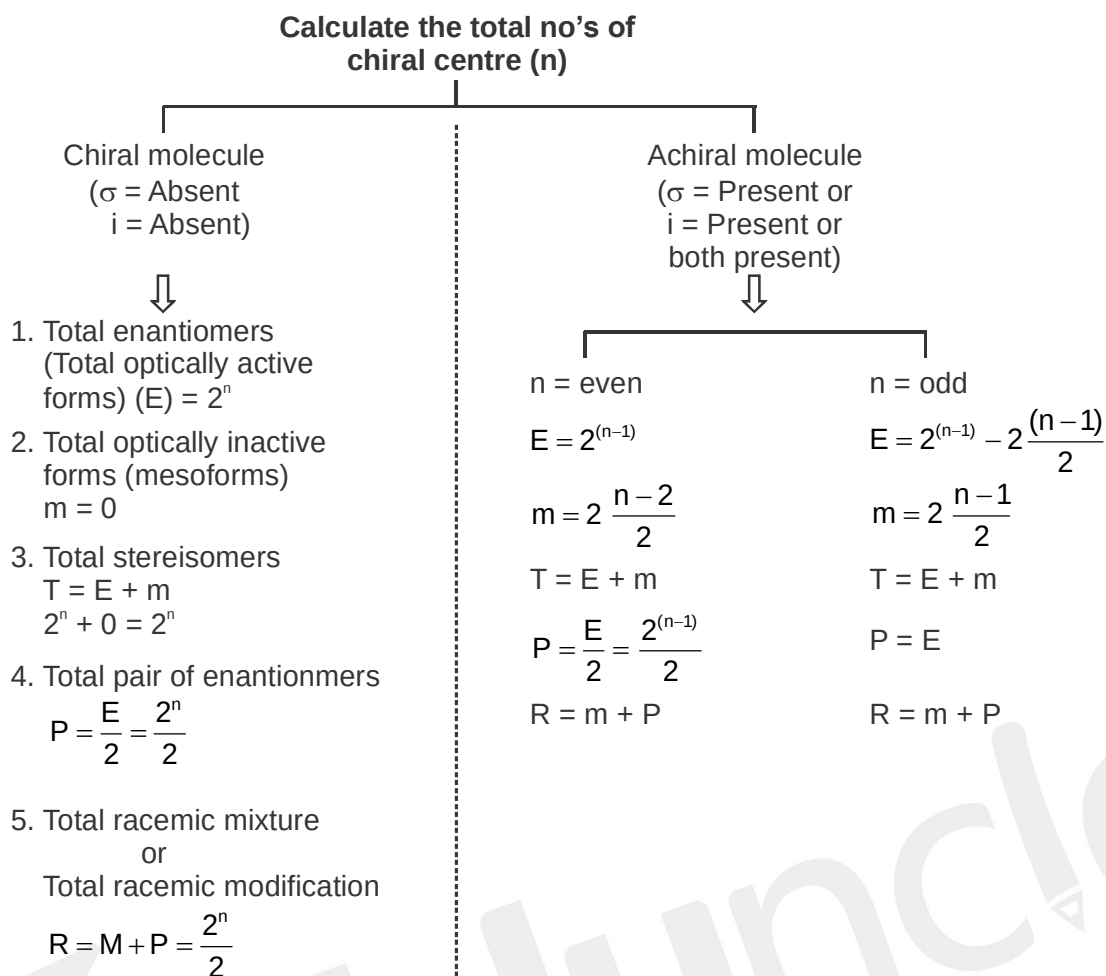


3. Trans cyclooctane :

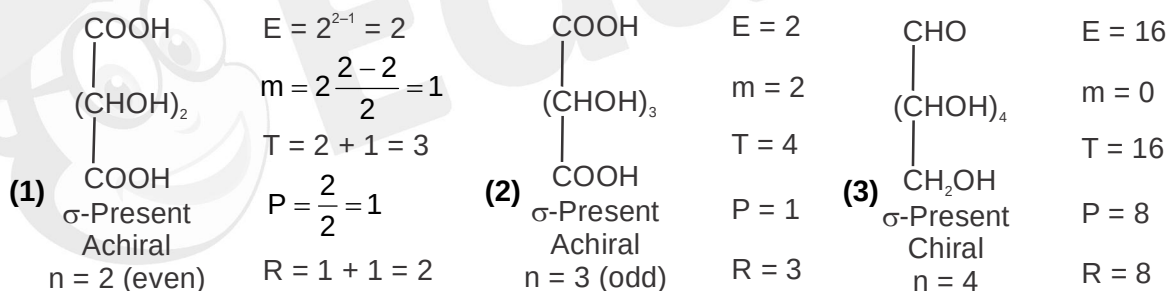


Calculation of optical isomers :

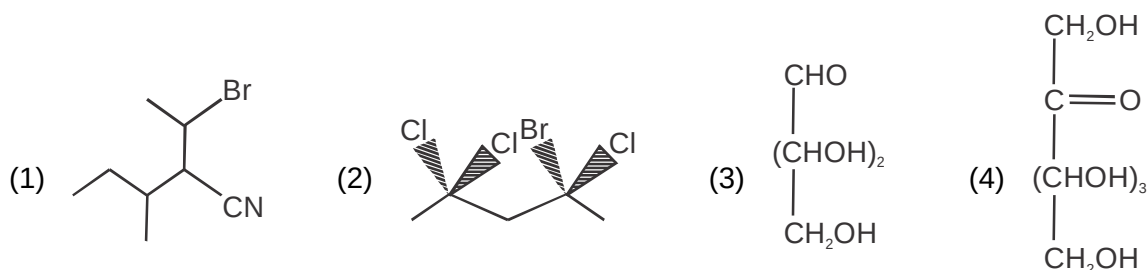
Optical isomer can be calculated by using following flow-chart



Example :



Que. Calculate the value of total optically active isomers ?

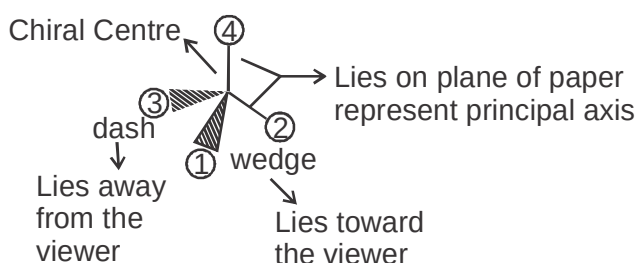


Ans. (1) 9, (2) 2, (3) 4, (4) 9

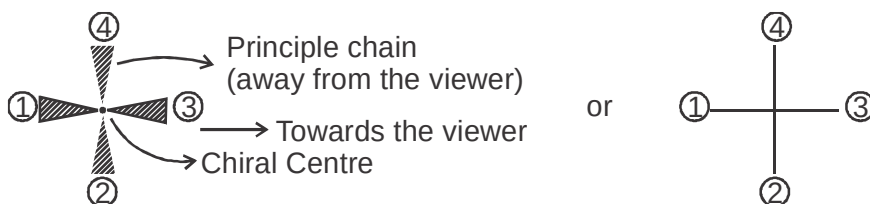
Configuration : The arrangements of atoms that characterizes a particular stereoisomer is called its configuration.

A. Representation of a configuration

1. Wedge-dash Representation :



2. Fischer Projection Formula :

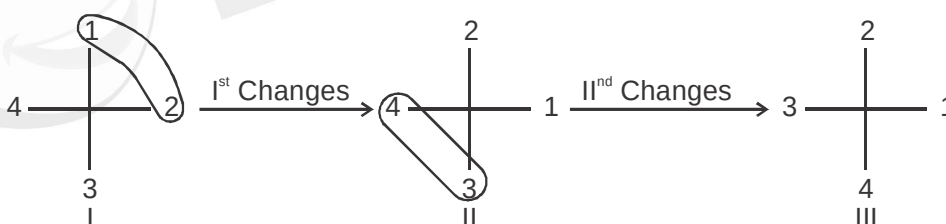


Characteristics

1. Fischer projection formula is always eclipsed conformation.



2. One inter change or odd number interchange of ligands at one chiral centre, gives enantiomers
Two interchange or even number interchange of ligands at one chiral centre, gives identical configuration.

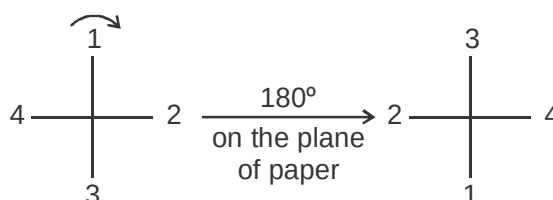


Ist and IInd = Enantiomers

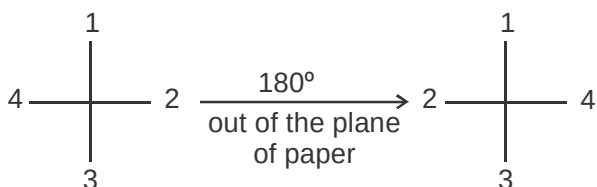
IInd and IIIrd = Enantiomers

Ist and IIIrd = Identical

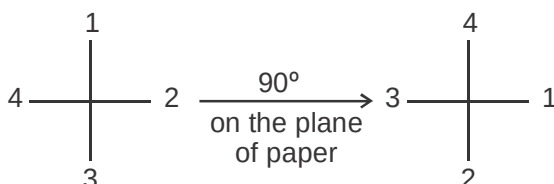
3. When fischer projection formula is rotated 180° on the plane of paper then it gives identical configurations.



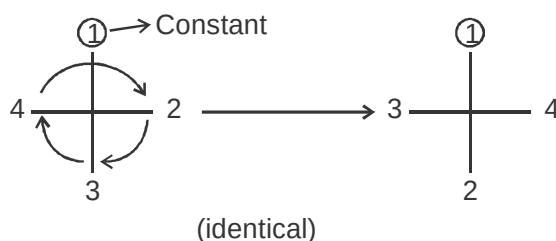
4. When fischer projection formula is rotated 180° out of the plane of paper, then it gives enantiomers.



5. When fischer projection formula is rotated 90° on the plane of paper then it gives enantiomers.

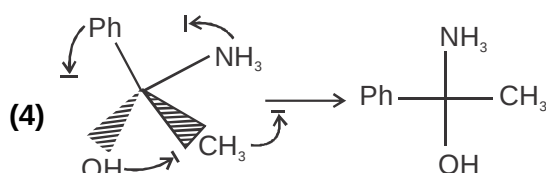
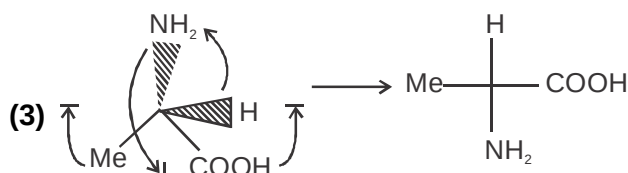
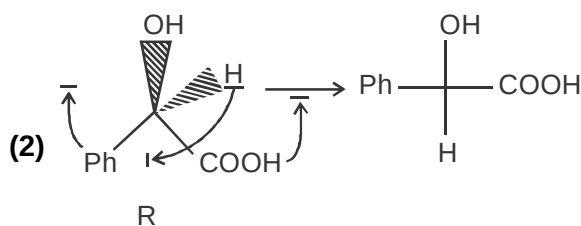
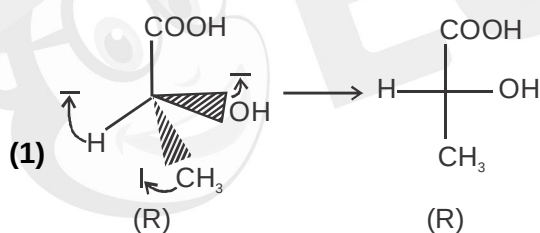


6.



Inter Conversion of Wedge-dash and Fischer Projection Formula :

The group which is below the plane of the paper (dash) is kept at the bottom and remaining three groups are attached in the same sequence (i.e., clockwise or anticlockwise) as they appear in the flying Wedge formula.



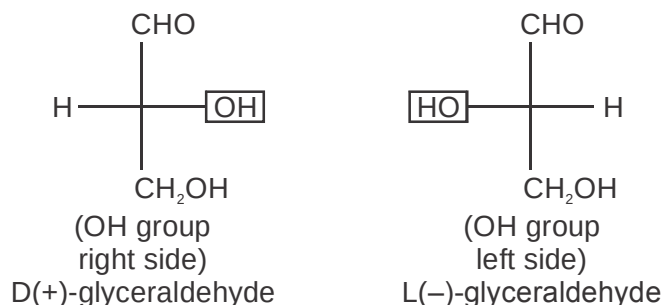
Assignment of Configuration :

1. D/L System (Relative configuration)
2. R/S Configuration (Absolute configuration)

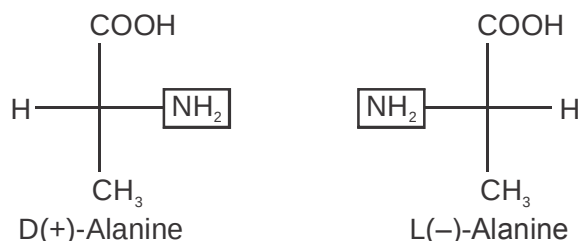
1. D/L System (Relative configuration) : In relative configuration we deal only with the configuration of one chiral Centre.

This method is used to assign the configuration of sugars and amino acids.

In the case of sugars the reference substances is glyceraldehyde.



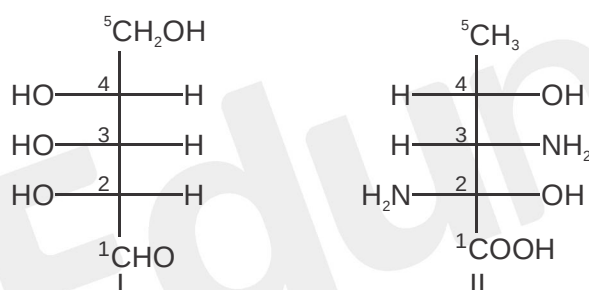
While in the case of amino acids the reference substances is alanine.



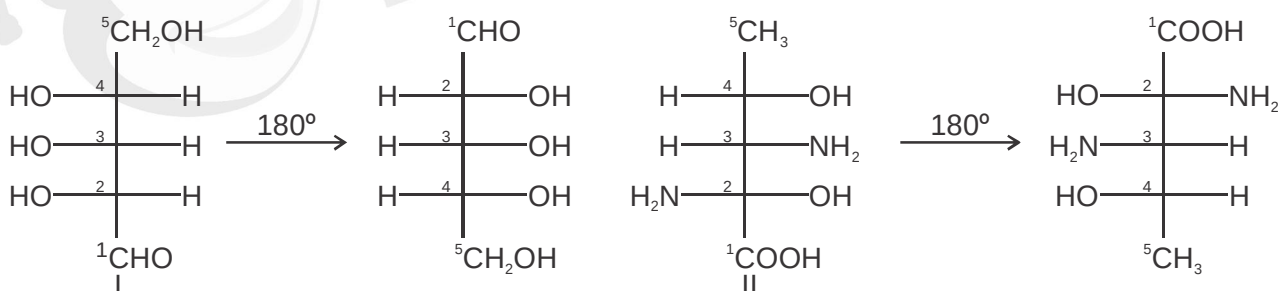
Rules for Determining Relative Configuration

Rule 1 : First the main chain determined and its atoms are numbered.

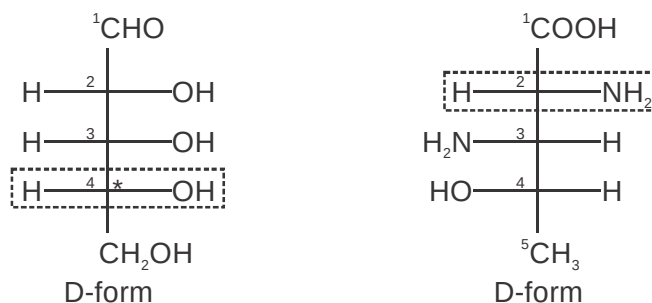
Example :

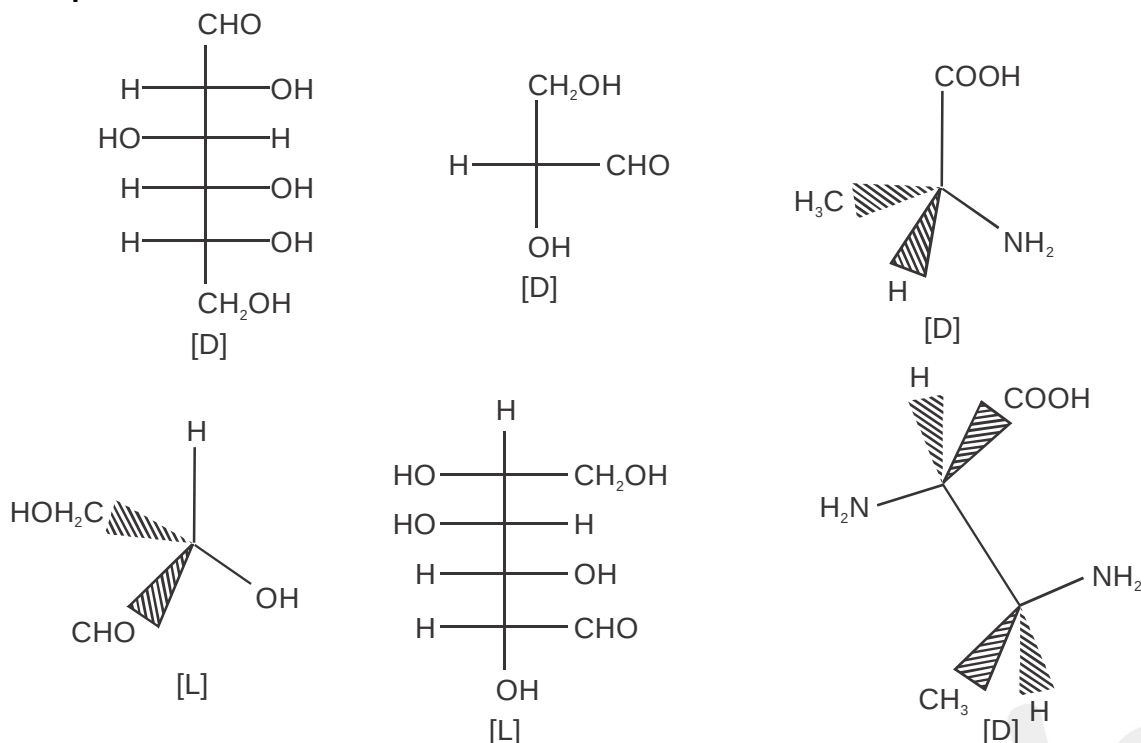


Rule 2 : Numbering should move from up to down. If it moves from down to up then we rotate the projection formula about 180° on the plane of paper.



Rule 3 : In the case of sugars; we will see the highest numbered chiral centres while in the case of Amino Acids, we will see the configuration of lowest no. of chiral centres



Examples :


2. R/S Configuration : To assign an enantiomer as R or S, first assign a priority (1, 2, 3 or 4) to each group

Priorities are assigned to groups by a series of rules known as the Cahn-Ingold-Prelog (CIP) sequence rule, named after the three chemists who developed them.

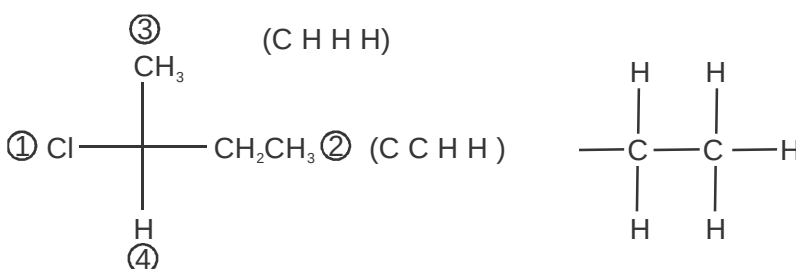
Rule 1 : The atom of highest atomic number gets the highest priority.



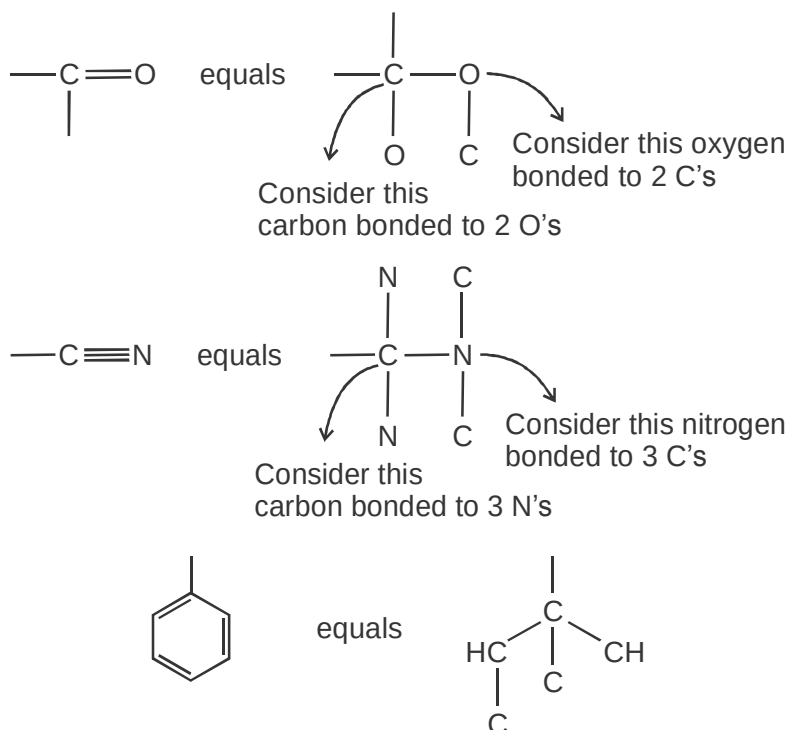
Rule 2 : If two atoms are isotopes of the same element, the atom of higher mass numbered has the highest priority.



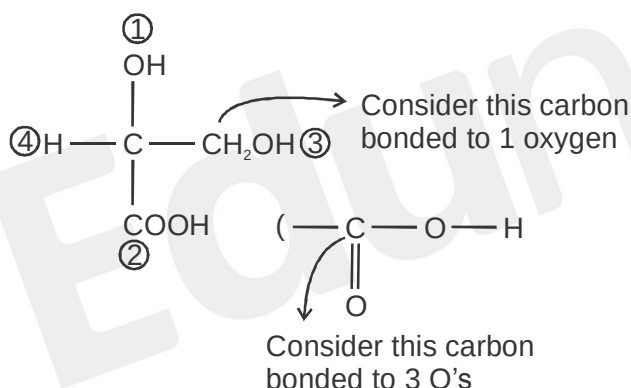
Rule 3 : When first atom is same then the next sets of atoms are examined. The process is continued.



Rule 4 : Groups containing double or triple bond are assigned as.



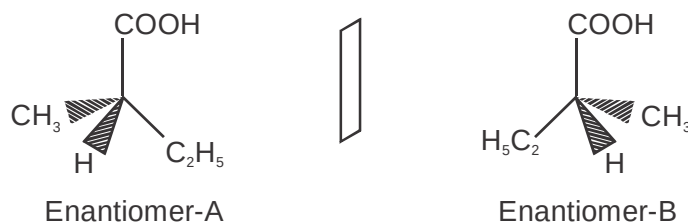
Example :



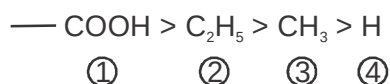
Assignment of R or S to a Chiral Centre

(A) R/S configuration in Wedge-Dash formula :

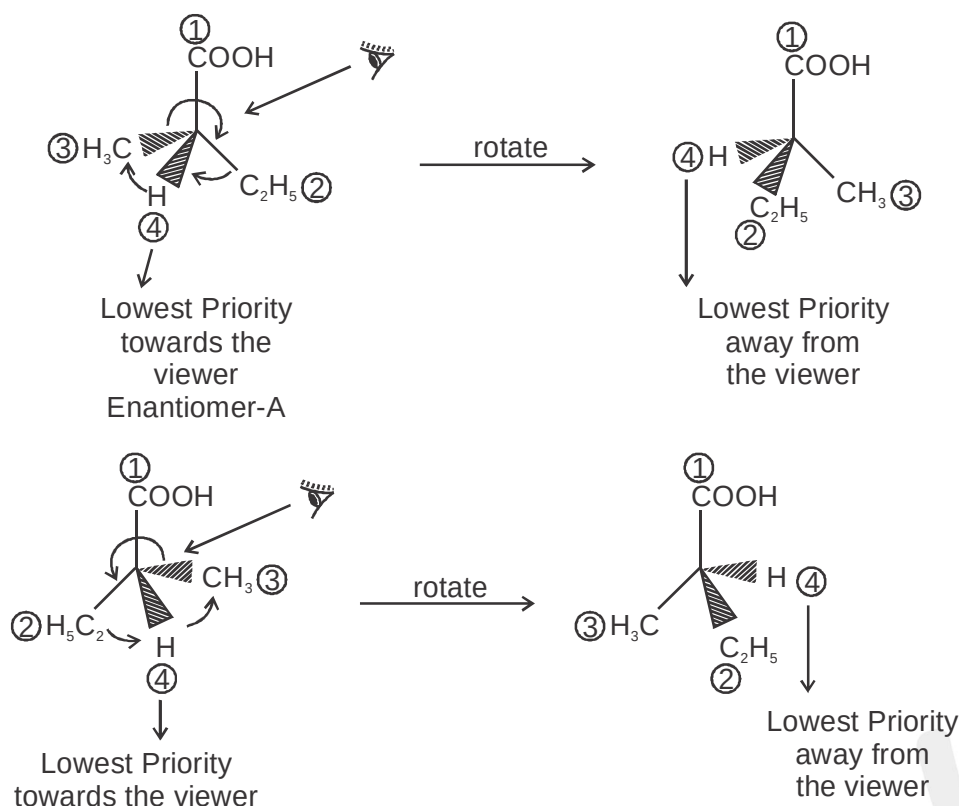
Example :



Step 1 : Assign a sequence of priorities 1 to 4 to the four atoms or group of atoms bonded to the chiral centre.

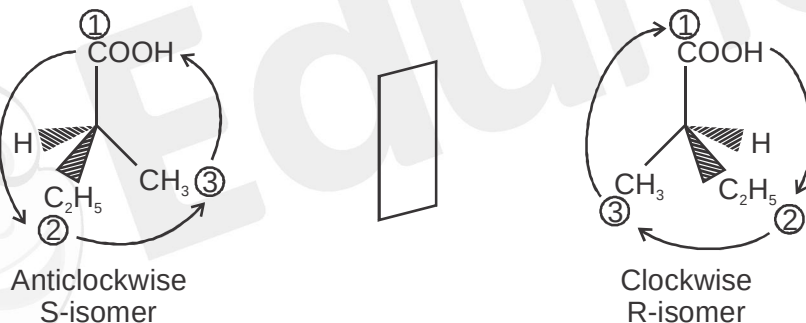


Step 2 : We visualize the molecule oriented so that the ligand/atoms of lowest priority is directed away from the viewer (on the dash)

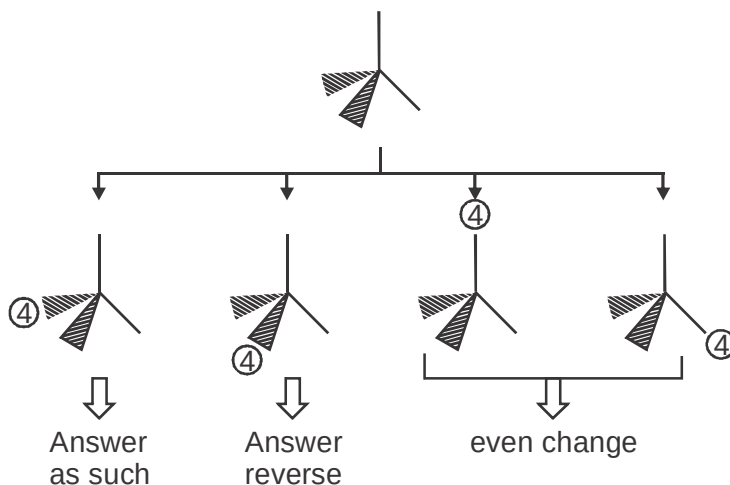


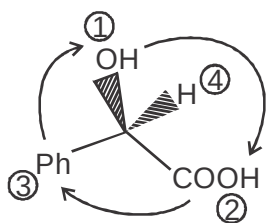
Step 3 : Trace a circle from priority group 1 $\rightarrow$ 2 $\rightarrow$ 3. If the arrow point of the circle goes in the clockwise direction, the configuration is specified R (Latin : Ractus = right).

If Anticlockwise the configuration is specified S (Latin : Sinister = left).

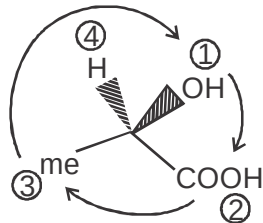


Simple method for Assignment of R or S

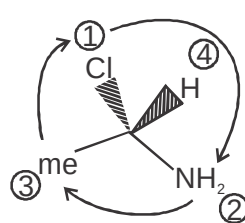


Examples :


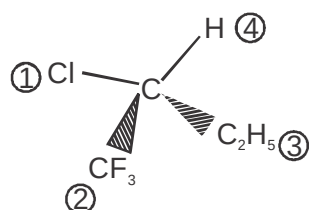
Lowest Priority
away from the
viewer
Answer as such
R-Configuration



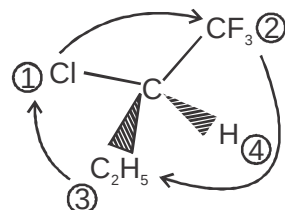
Lowest Priority
away from the
viewer
Answer as such
R-Configuration



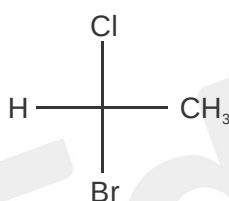
Lowest Priority
towards the viewer
Answer reverse
S-Configuration



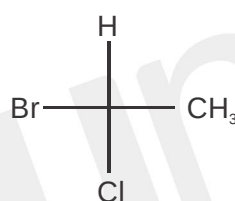
Even
Change



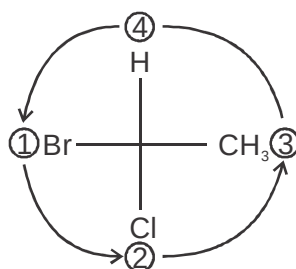
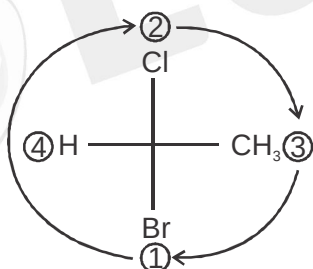
R-Configuration

R and S configuration in fischer projection
Example :


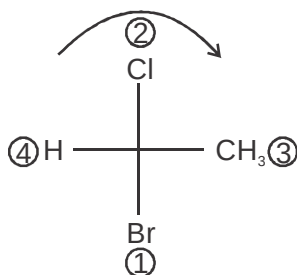
1-bromo-1-chloro-ethane



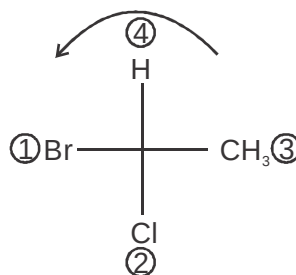
Step 1 : Assign a sequence of priority 1 to 4 and trace a circle from priority group 1→2→3→4.



Step 2 : Case-I if the group of lowest priority is on the horizontal line then clockwise sequence gives S and Anticlockwise sequence R configuration.

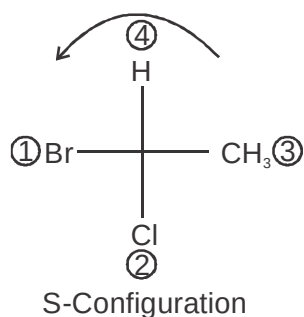


Lowest Priority left side
clockwise
[S]-Configuration

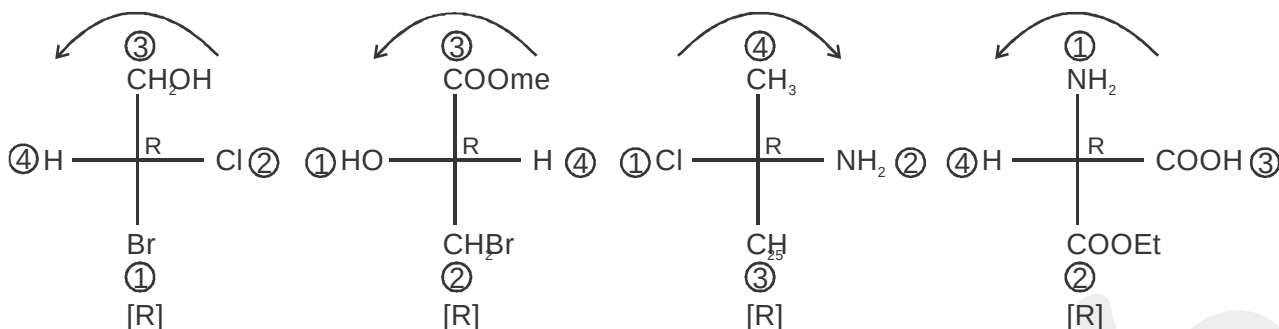


Lowest Priority is
above the plane
Anticlockwise

Step 3 : Case-II If the group of lowest priority is above or below the plane then clockwise sequence gives R and anticlockwise sequence S configuration.



Examples :



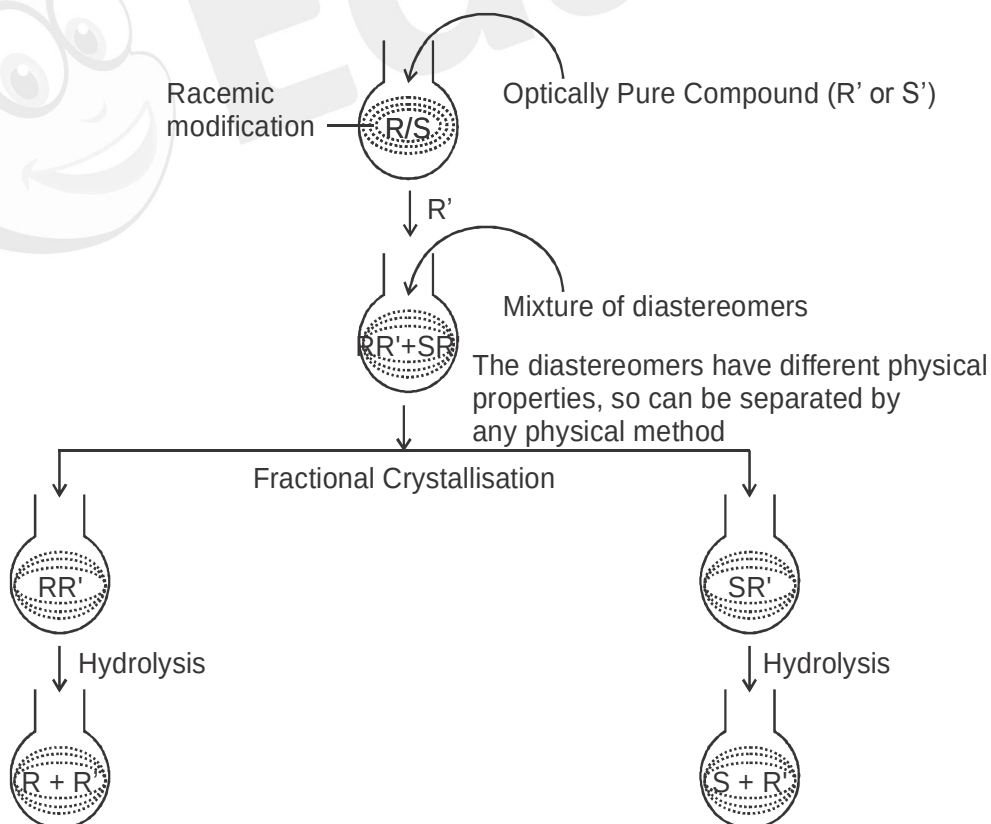
Resolution :

The Separation of a racemic modification into its constituent enantiomers is known as resolution.

Enantiomers have identical, physical and chemical properties so these cannot be separated by normal physical method.

Separation of pair of enantiomer by using Chemical method (Via Diastereomers)

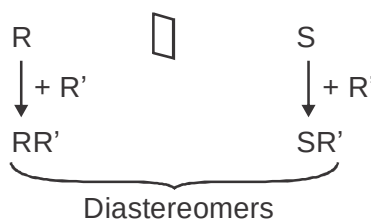
This is the best method of resolution. In this method a racemic modification is converted into a mixture of diastereomers by using a pure enantiomer of another compound.



Properties of Enantiomers

Enantiomers have identical physical and chemical properties except :

1. Odour
Ex. (+) – Limonene has smell of oranges and
(–) – Limonene has smell of lemons
2. Optical rotation (equal magnitude but opposite direction)
3. Physical activity
Ex. (–) – Nicotine is poisonous than (+) – Nicotine;
(+) – histidine is sweet whereas (–) – histidine is tasteless.
4. Reactivity towards optically active compound because it causes them to behaves as Diastereomers.



Diastereomers have different (optically active) physical properties

Diastereomers react with O.A (optically active) as well as optically inactive (OI) compounds with different rate.

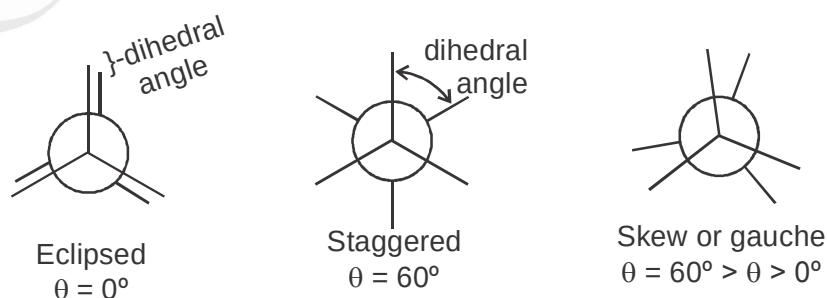
Conformational Analysis

Special orientation of molecule which differ only in the dihedral angle and easily inter convertible by rotation about single bond are called conformers. They are also known as rotational isomers or rotamers a molecule may have a number of conformers each different conformation will have a different energy.

Conformational analysis is the analysis of physical and chemical properties of a compound in term of various conformations in ground state, transition state and also excited state.

Dihedral Angle

The angle that separates a bond on atom from a bond on an adjacent atom is called a dihedral angle. There are three conformers are possible.



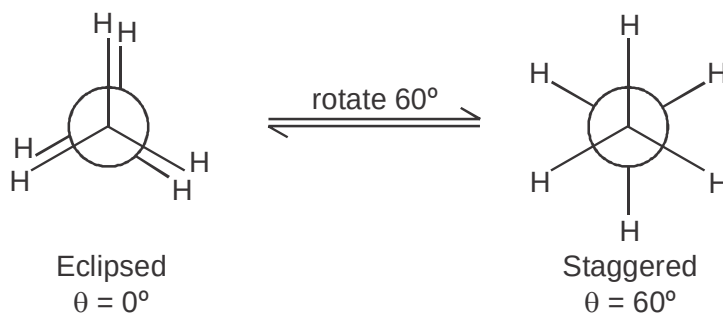
4. CONFORMATIONS OF ACYCLIC SYSTEM

Conformations of Ethane

When ethane molecule rotates about carbon-carbon single bond.

Two extreme conformations can result

1. Staggered conformation
2. Eclipsed conformation



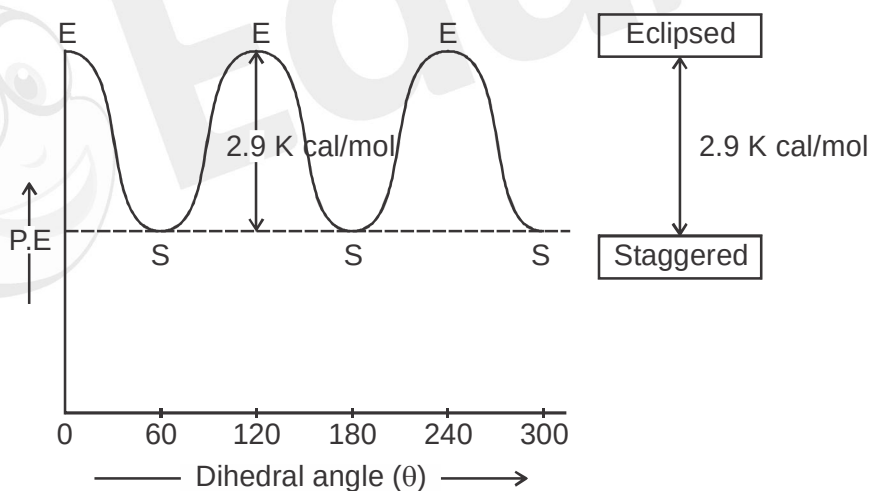
(Newman Projection of Ethane)

The staggered and eclipsed conformations of ethane interconvert at room temperature. The staggered conformations are more stable (lower in energy) than the eclipsed conformations.

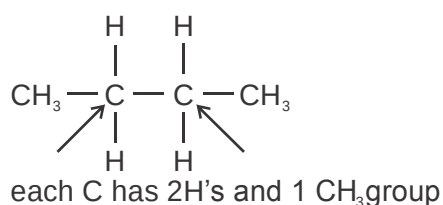
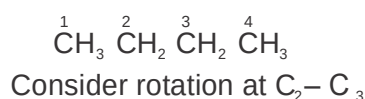
Reason : In eclipsed conformation carbon-hydrogen bonds are closest so electron-electron repulsion between the bonds in the eclipsed conformation increases its energy compared to the staggered conformation.

The potential energy barrier in ethane is 2.9 kcal/mol. The extra energy of the eclipsed conformation is called torsional strain.

Energy profile diagram of conformations of ethane :

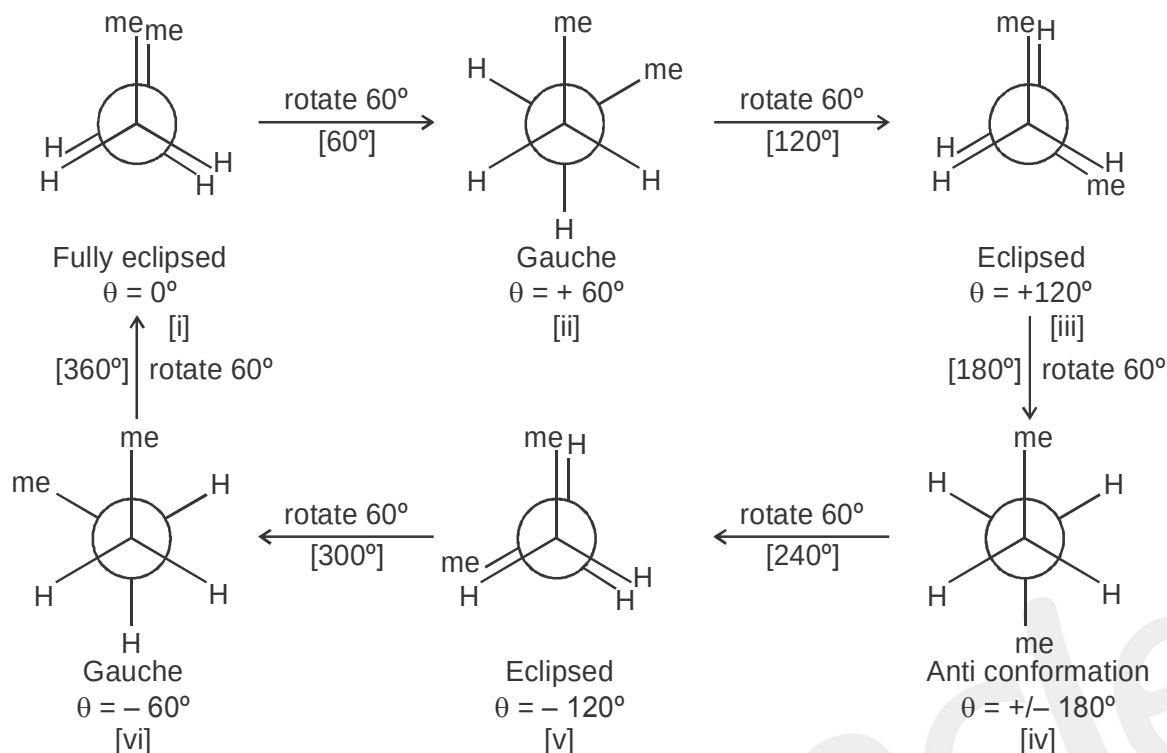


Conformations of Butane :



In the case of n-butane rotation about C_2-C_3 bond gives rise to six distinct conformations as shown below.

The conformations are raised out of successive rotations by a dihedral angle of 60° each time.

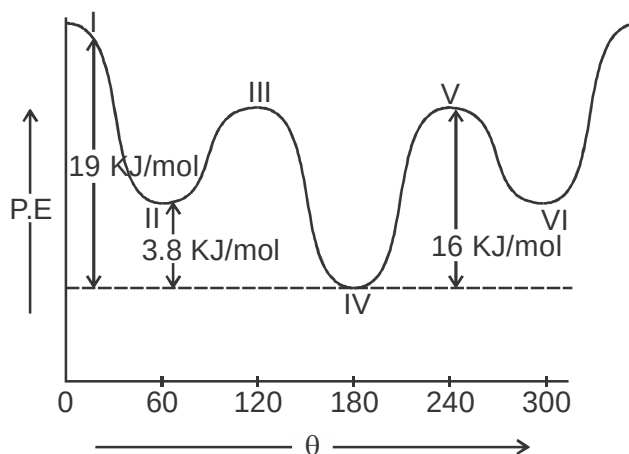


- The staggered conformation (II, IV, VI) are lower in energy (more stable) than the eclipsed conformations (I, III, V)
- The $-CH_3$ groups are further apart in the anti conformation (IV), than the gauche conformation (II, VI) so among the staggered conformations IV is lower in energy (more stable) than II and VI
- Conformation I is higher in energy than III or V because two longer CH_3 groups are force close to each other.

Over all potential energy = $I > III \approx V > VI \approx II > IV$

Stability = $IV > II \approx VI > V \approx III > I$

Energy Profile Diagram of Conformations of Butane



5. CONFORMATIONS OF CYCLOHEXANES

For Cyclohexanes there are two extreme conformations

(A) Chair Conformation : A planar cyclohexane would experience angle strain (bond angle is different from desired tetrahedral bond angle of 109.5°) and torsional strain (caused by repulsion between bonding electrons)



So in order to minimize above repulsion, cyclohexane adopts a puckered conformation called the chair form which is more stable than any other possible conformation.

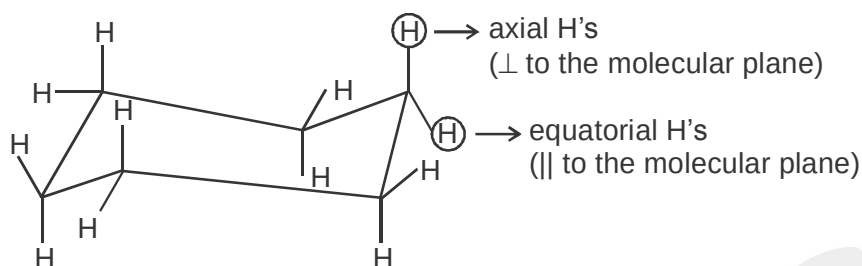
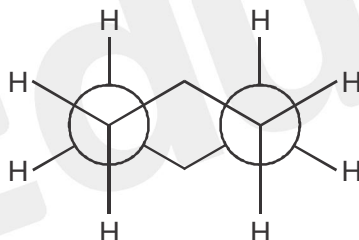


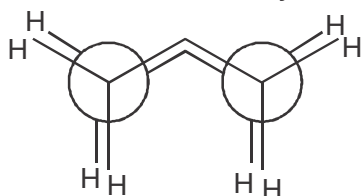
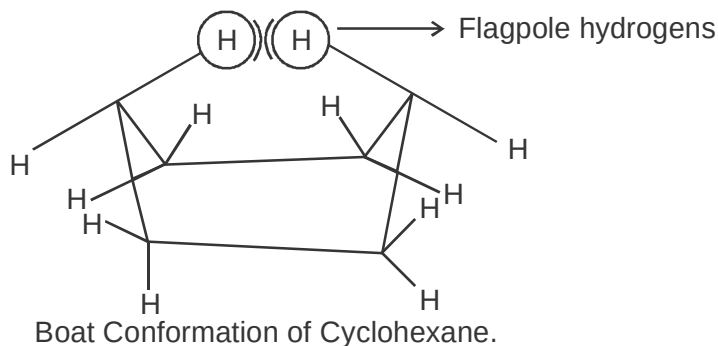
Fig. : Chair Conformation of Cyclohexane

- Cyclohexanes has six axial H's and six equatorial H's
- It is a staggered conformation



(Newman Projection of the Chair Conformation)

(B) Boat Conformation : It is a eclipsed conformation of cyclohexane. It is less stable than the chair conformation due to torsional strain furthermore in the boat form there is steric repulsion between the two hydrogens (flagpole hydrogens)



Newman Projection of the Boat Conformation

Ring Flipping

At room temperature cyclohexane rapidly interconvert (flip) to mirror image chair conformations.



Axial and equatorial H atoms are interconverted during a ring flip. Axial H atoms become equatorial and equatorial H atoms become axial H atoms.

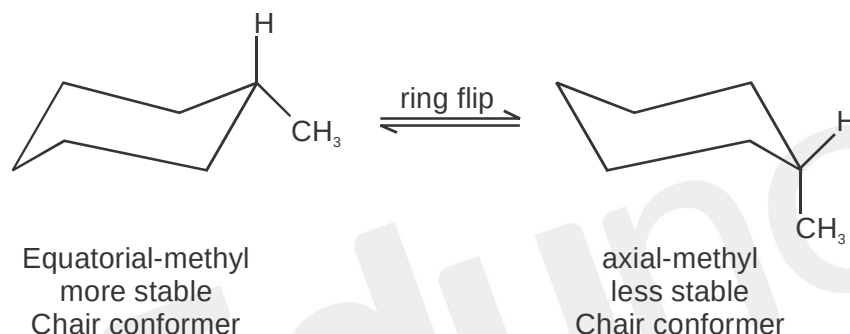
Conformations of Monosubstituted Cyclohexanes

When one hydrogen in cyclohexane is replaced by a larger atom or group, crowding occurs.

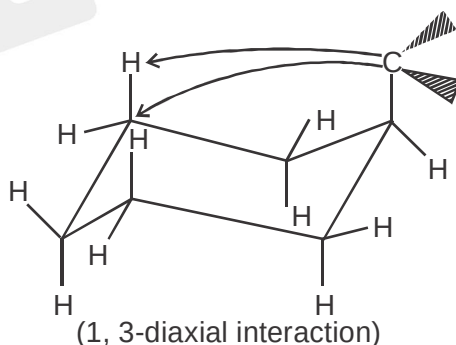
Ex. methyl cyclohexane

The two chair conformers of a monosubstituted cyclohexane such as methyl cyclohexane are not equivalent.

The methyl substituent is in an equatorial position in one conformer and is in an axial position in the other.



The chair conformer with the methyl substituent in an equatorial position is the more stable because a substituent has more room and therefore steric interaction. These steric interactions are called 1,3-diaxial interaction.



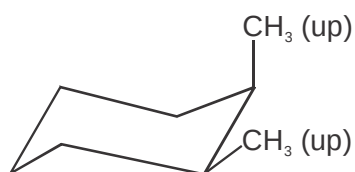
Conformations of disubstituted Cyclohexanes :

Rotation around the C-C bonds in the ring of a cyclohexane is restricted. So disubstituted Cyclohexanes give geometrical isomerism.

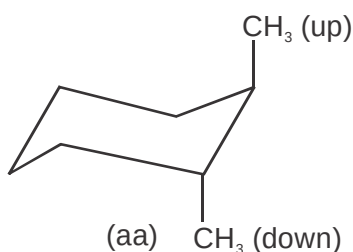
One has both methyl substituents on the same side of the cyclohexane ring (up-up or down-down); it is called the cis isomer.

The other has the two methyl substituents on opposite sides of the ring (up-down), it is called the trans isomer.

(1) 1, 2-dimethyl Cyclohexane :



(ae or ea)
Cis isomer
chiral
but optically
inactive due to
conformational
enantiomerism

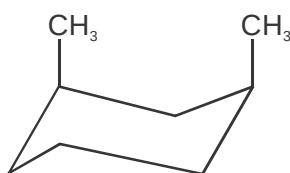


(aa) CH₃ (down)
Trans isomer
 σ = Absent
i = Absent
chiral
optically
active

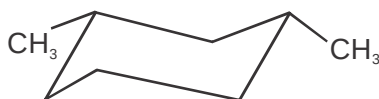


(ee)
Trans isomer
 σ = Absent
i = Absent
chiral
optically
active

(2) 1, 3-dimethyl Cyclohexane :



(aa)
Cis isomer
 σ = Present
Achiral
optically inactive



(ee)
Cis isomer
 σ = Absent
chiral
optically active

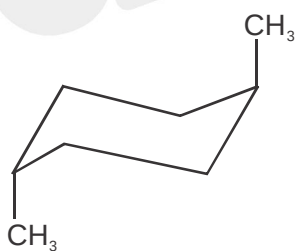


(a, e)
Trans isomer
 σ = Absent
chiral
optically active

(3) 1, 4-dimethyl Cyclohexane :



a, e or e, a
Cis isomer
Plane (σ) = Present
Achiral
optically inactive



a a
Trans isomer
Plane (σ) = Present
Achiral
optically inactive



e e
Trans isomer
Plane (σ) = Present
Achiral
optically inactive